

Evidence-Monotone Audio Representation: Preventing Provenance Promotion Across Derived Audio Transformations

Alejandro Urias^{a,*}

^a*Independent researcher, Hermosillo, Sonora, Mexico*

Abstract

Derived audio can keep its waveform while strengthening what it says about its origin. Under a boundary-only inheritance policy, a constructed reference and not any named product, a clip assembled from captured and generated material, trimmed, resampled and re-encoded, emerges described as captured-only. We specify an operator contract for derived audio: over the complete required source set, provenance unions, same-channel support minimises, applicability intersects and lineage unions, and an unverified source interval is absorbing. The algebra is a known meet-semilattice construction; new are its executable temporal instantiation with declared kernel footprints, a signed transport inside C2PA temporal regions, and the evaluation. Across 984 213 contract checks, 10 000 adversarial timelines, 600 mixed-origin clips transformed by stock FFmpeg and 10 860 differential-oracle cases, complete-source propagation produced 0 failures and 0 promotions, whereas boundary-only inheritance promoted in 3 600 of 4 800 stock-FFmpeg transformation runs. Decoded essence was identical before and after signing in 120 round-trips, at $3.37\times$ the bookkeeping baseline. An independent reproduction on other hardware and FFmpeg major version returned every conformance, promotion and oracle outcome unchanged and found two build-specific declarations its build does not satisfy, and a holdout challenge exceeded the measured encoder and stretcher declarations on the reference build: the algebra holds by proof, the audio instantiation only where the declared map contains the real dependencies. The contract constrains what a derived representation may claim about its sources, and binds only an operator that implements it; it does not detect deepfakes, and a valid signature proves only that an assertion was signed.

Keywords: digital evidence, media provenance, content credentials, C2PA, audio forensics, derived media, reproducibility

1. Introduction

Speech can now be generated convincingly enough that “is this recording real?” is answered less and less reliably by listening. Detection asks whether a signal carries the traces of synthesis (Pham et al., 2025; Wang et al., 2026); provenance asks what an asset’s own signed history says about where it came from (Coalition for Content Provenance and Authenticity, 2026a). This work is about the second question, so nothing here is a detection result.

*Corresponding author. ORCID: 0009-0007-3147-323X.
Email address: alex@aurtech.mx (Alejandro Urias)

Provenance systems bind assertions to an asset. They do not, on their own, say what a *derived* asset is entitled to assert, and derivation is routine: a clip is trimmed, spliced, resampled and re-encoded before it reaches an examiner, and each step writes a new manifest. The C2PA specification names the relationship of such an asset to its input `parentOf` (Coalition for Content Provenance and Authenticity, 2026a, Table 10, §18.16.3), and its vocabulary can carry everything the problem needs: `componentOf` ingredients with their own histories, a `digitalSourceType` for generative origin, assertions scoped to a temporal range. What the text does not prescribe, and we searched it for such a rule, is the *reduction*: the aggregate evidence state a derived interval should emit when the component histories it represents disagree. The coalition’s July 2026 guidance on identifying synthetic content develops the vocabulary for declaring AI involvement and localising it to regions (Coalition for Content Provenance and Authenticity, 2026b); that is guidance on how to *represent* what happened to part of an asset, and it is complementary to what this paper supplies, the rule for what a derived interval may *claim* once its parts disagree.

Consider five consecutive source intervals of a recording whose third is generated speech. A derived asset representing all five can be described from its retained boundaries, which are captured, and emerge labelled captured-only; the decoded audio is bit-identical either way (Figure 1). An examiner reading that claim is being told something stronger than the sources support, by a chain in which no signature was forged, and wherever a disclosure obligation is discharged by reading a manifest rather than inspecting a signal the result is not a missed detection but a confident wrong answer with a valid signature attached.

We call this *provenance promotion*: a derived representation acquiring a stronger evidential claim than the complete source material it represents, without any change to the decoded signal. The requirement that prevents it, *evidence monotonicity*, is that along a chain of representation-only transformations the emitted claim is no stronger than the meet of every required source, ancestry accumulates rather than disappears, and the absence of verified evidence is absorbing.

Digital forensics already treats the step from stored evidence to software-derived representation as a place where meaning can change, and two failure modes recur: what a representation aggregates away (Hargreaves et al., 2024; Andersen et al., 2025), and how far the same nominal input is read differently by different builds and versions (Nordvik and Axelsson, 2022; Külper et al., 2026). Derived-media provenance meets both at once: heterogeneous source evidence has to be reduced to one claim, and the processing dependencies that decide which sources count are fixed by the build, so a conservative reduction rule and validated dependency declarations are both needed.

An earlier instantiation of that requirement exists for spatial data, where derived terrain geometry can overstate the evidence of the source arc it replaces (Urias, 2026). This work asks whether the requirement is usable in a temporal medium with a real signed transport and real processing software, and what it costs. What is inherited and what is new, row by row, is Supplementary Note S1, and no earlier result is reused: the algebra transfers, and the temporal interval maps, the kernel-footprint problem, the validations against third-party processing, signal transparency and the signed transport are not part of the spatial work.

The algebra, a meet over a semilattice of labels with union of ancestry and an absorbing bottom element, is an instance of constructions that are decades old, in semiring-based provenance annotation (Green et al., 2007) and in label lattices for secure information flow (Denning, 1976; Goguen and Meseguer, 1982). We do not claim the non-amplification property as a new theorem; we state it as an instance, prove the footprint-monotonicity result the temporal instantiation relies on, and locate the contribution in what is executable and measured.

Contributions. Each is traceable to a completed experiment.

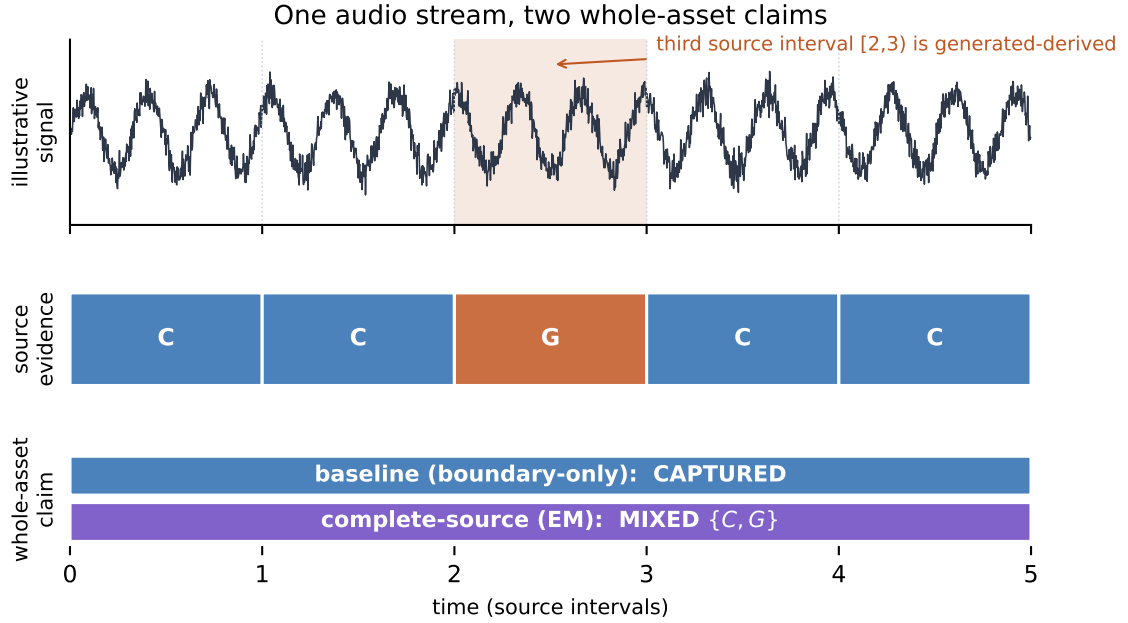
1. **The contract, and its temporal instantiation.** An operator contract for derived media stated without reference to any modality, its non-amplification property presented as an instance of a known meet-semilattice construction (Section 4); instantiated over sample-exact half-open intervals for eight representation operators, each with a declared conservative kernel footprint, with a footprint-monotonicity result showing that over-approximating the required source set is always safe (Section 5, Proposition 2).
2. **A signed transport, and an ablation of the rule.** Interval evidence carried inside C2PA temporal regions of interest, surviving derivation and re-signing on 3 containers with 120 round-trips, with a C2PA-native composition of signed captured and generated `componentOf` ingredients (Sections 5.4, 7.5, 7.8); and a policy ablation showing that complete-source aggregation and declared kernel footprints are each necessary and neither sufficient (Section 7.3).
3. **Evidence that it holds, and what it costs.** An executable conformance suite of 984 213 checks over every source word of length up to 8 over $\{C, G, \perp\}$ with 0 failures, and a two-language differential over 10 860 frozen cases with 0 disagreements, those frozen cases being the conformance target a third-party reimplementation can be run against (Sections 7.1, 7.10); 600 mixed-origin clips processed entirely by stock FFmpeg, with the interval mapping validated against software we do not control (Section 7.4) and the declared footprints tested separately by differential influence probes (Section 7.6), since agreement on output length says nothing about how far a dependency reaches, both passes packaged as `tools/calibrate_footprint.py`, then challenged on held-out signals, which the encoder and stretcher declarations fail (Section 7.7); and cost on both axes, runtime, where bookkeeping is 0.114% of the processing it accompanies and the assertion grows at 333 B per evidence interval (Section 7.11), and *evidential*, a claim-dilution metric priced in samples per ancestry boundary rather than in fractions of the asset and compared with a whole-asset claim (Section 7.9).

The contract does not infer authenticity from acoustic features; it prevents a provenance-aware chain from promoting a mixed or unverified source interval into a stronger captured-only class, and Section 3 states what it does when evidence is absent. It is orthogonal to passive deepfake detectors and to watermarking, and composes with either. Transparency obligations under Article 50 of Regulation (EU) 2024/1689 became applicable on 2 August 2026 (European Parliament and Council of the European Union, 2024), which makes machine-readable origin claims about derived media operationally relevant; we make no legal claim about what any disclosure regime requires.

2. Related work

Detection of synthetic speech. Anti-spoofing and deepfake-speech detection classify a signal by its acoustic traces, and lose accuracy under unseen attacks and codecs, as documented (Wang et al., 2026; Pham et al., 2025). That line guarantees a decision about a signal; it does not constrain what a derived asset may assert about its own sources, and it needs no source evidence to exist.

Watermarking and soft bindings. Localised audio watermarking marks generated speech at synthesis time and detects the mark per region (San Roman et al., 2024); C2PA treats watermarks and fingerprints as *soft bindings* that recover a manifest separated from its asset (Coalition for Content Provenance and Authenticity, 2026a, §18.10). Neither defines the aggregate claim of a derived asset whose sources differ, and a region carrying no mark is not thereby captured.



Illustrative signal, not a corpus clip. The waveform is identical under both policies; only the evidential claim differs.

Figure 1: The minimal temporal counterexample. Five source intervals, of which the third is generated-derived. A boundary-only reading of the retained region sees captured evidence at both ends and may emit CAPTURED; the complete-source rule sees $C + C + G + C + C$ and emits MIXED $\{C, G\}$. The decoded audio is identical under both policies.

120 *Cryptographic media provenance.* C2PA binds signed assertions and transformation history to
 121 an asset, defines the ingredient relationships `parentOf`, `componentOf` and `inputTo`, and scopes
 122 assertions to temporal regions of interest ([Coalition for Content Provenance and Authenticity, 2026a](#), §18.2.2.3, §18.16.3). Security analyses report problems in the cryptographic and trust
 123 layers ([Golaszewski et al., 2026](#)), and related work constructs assets whose manifest and watermark
 124 contradict each other while each passes its own verification ([Nemecek et al., 2026](#)). Our concern is a
 125 different layer, orthogonal to both: assuming keys are intact, signatures verify and no assertion is
 126 omitted in bad faith, is the assertion a conforming derived asset emits conservative with respect to
 127 every source interval it represents? The specification preserves a great deal, since a `componentOf`
 128 ingredient carries its own manifest and a generative component its `digitalSourceType` ([Coalition
 129 for Content Provenance and Authenticity, 2026a](#), §18.16.3), and a consumer walking the full graph
 130 could reconstruct everything. What version 2.4 does not define is the aggregate: no normative rule
 131 states that a derived interval’s own claim must be the conservative reduction over all its components
 132 rather than an inheritance from its parent. Our composition experiment is the evidence: every
 133 composition carries two ingredients whose declared source types disagree, and the official validator
 134 reports each as valid without computing or requiring any aggregate over them (Section 7.8).
 135

136 *Inferring audio provenance from the signal.* *Audio provenance analysis* reconstructs content reuse
 137 and parent-child relations across a heterogeneous collection with no prior knowledge of its contents,
 138 combining partial matching with phylogeny ([Gerhardt et al., 2024](#)). That line infers a lineage where
 139 none was recorded; the contract here assumes signed source evidence exists and constrains how a
 140 known operator may aggregate it. A pipeline could reasonably use the first to recover a lineage and

the second to decide what the recovered lineage entitles a derived interval to claim.

Provenance for edited audio. Chunk-localised speech provenance binds short segments to an issuer-signed original through perceptual fingerprints and Merkle commitments, addressing the fragility of hard bindings under re-encoding (Ono, 2026); zero-knowledge constructions prove an edit was drawn from an allowed set while hiding source, edits and device (Fu et al., 2026). Neither states an aggregation rule for when the represented sources disagree, and neither proves non-amplification.

Provenance models and information flow. W3C PROV records derivation as a graph (World Wide Web Consortium, 2013): a PROV graph for the five-interval counterexample would hold `wasDerivedFrom` edges to all five sources, a complete and true record, and still say nothing about which *claim* the derived segment may emit. The machinery for constraining that reading is established: provenance semirings annotate derived data so that combination is associative and distributes over alternative derivations (Green et al., 2007), though idempotence, which the meet here has, is not part of that framework and is incompatible with several of its instances; information-flow lattices require combination to move only in the conservative direction (Denning, 1976; Goguen and Meseguer, 1982). The property we need is the meet in such a structure, and we present it as such.

Forensic representation and tool-dependent interpretation. Hargreaves et al. (2024) model a forensic tool as connected processing stages in which an error introduced at one stage propagates downstream unless intermediate outputs are exposed and validated, and empirical work fills in what that costs. Andersen et al. (2025) reported examiner errors associated with misleading presentation in the semi-automated tools used in that study, Cellebrite and APOLLO, naming aggregation, missing detail and timestamp presentation among the contributing factors. Interpretation also varies by implementation rather than by specification: Nordvik and Axelsson (2022) measured substantial inconsistencies in exFAT timestamp handling across the operating systems and the four forensic tools they examined, among them Autopsy and EnCase Examiner, and Külper et al. (2026) found largely undocumented changes in exported report output across twenty-five versions of Autopsy. None of these studies concerns provenance or demonstrates promotion, and the products named in them appear here only as their subjects. The premise they establish is that preserving source data is not sufficient if the derived representation changes the evidential meaning an examiner sees.

What is missing. Table 1 places these lines side by side. No row provides an operator-level complete-source rule instantiated over media time and carried in a signed transport; that combination is what this work contributes, and the matrix behind the table, with search dates and query strings, is Supplementary Note S2.

Source attribution from the signal, such as identifying a recording microphone (Qamhan et al., 2023), infers origin from the audio, where this work reads no audio at all. A contract is worth nothing unless its implementation can be checked, which is why the conformance suite of Section 7.1 is written against the requirements rather than against our code.

3. Threat model

In scope. Provenance promotion and laundering through representation-only transformations, under four conditions: source evidence exists; the processing system holds the source-to-output mapping; the operator is expected to conform to the contract; and cryptographic primitives and signing keys are intact.

Table 1: Positioning. Each row names a guarantee the line already provides, then the evidence property it does not, by itself, enforce on a derived audio interval. The contribution here is the joint enforcement of the right-hand column through representation-only operators, not the reinvention of any left-hand guarantee.

Line of work	Established guarantee or finding	Left unenforced on a derived interval
Synthetic-speech detection (Wang et al., 2026; Pham et al., 2025)	A decision about a signal from its acoustic traces	What the derived asset may assert about its own sources
Localised watermarking and soft bindings (San Roman et al., 2024; Coalition for Content Provenance and Authenticity, 2026a)	Region-level presence of a mark; recovery of a separated manifest	Aggregation of heterogeneous source evidence into one claim
Cryptographic provenance (Coalition for Content Provenance and Authenticity, 2026a)	Signed assertions, ingredient relationships, temporal regions of interest	A rule requiring the derived claim to be conservative over all ingredients
C2PA security analysis and cross-layer attacks (Golaszewski et al., 2026; Nemecek et al., 2026)	Weaknesses in the trust, timestamp and validator layers; contradictions between layers	Behaviour of a chain in which every layer is intact and honest
Audio provenance analysis (Gerhardt et al., 2024)	Reconstruction of reuse and parent-child relations from unknown signals, via partial matching and phylogeny	What a derived interval may claim once its recorded evidence disagrees
Provenance for edited audio (Ono, 2026; Fu et al., 2026)	Chunk-to-original binding after transport; zero-knowledge edit correctness	An aggregation rule when represented sources disagree
Provenance graphs and lineage (World Wide Web Consortium, 2013)	A truthful record of derivation	Operator-level monotonicity as an executable contract
Provenance semirings and information-flow lattices (Green et al., 2007; Denning, 1976)	The algebra of conservative combination	A temporal media instantiation with a signed transport
Forensic tool representation and validation (Hargreaves et al., 2024; Andersen et al., 2025; Nordvik and Axelsson, 2022; Külper et al., 2026)	Evidence that representation and output can vary across processing stages, tools and versions	A conservative operator-level rule for what a derived interval may claim about heterogeneous source evidence
Verifiable chain-of-custody ledgers (Burri et al., 2020)	That a record of handling is complete and unaltered	What a derived asset may claim about material it represents

Out of scope. Capture devices that lie about what they recorded; compromised signing keys; dishonest certificate authorities; unsigned media of unknown origin; speaker identity; the truthfulness of what is said; semantic misinformation; waveform-artefact deepfake detection; watermark removal; and full provenance stripping followed by construction of a fresh, dishonest trust chain. Attacks on the

cryptographic and trust layers are the subject of separate analyses (Golaszewski et al., 2026) and are assumed away here precisely so that the residual semantic question is isolated.

Required behaviour when evidence is absent or invalid. The state is UNVERIFIED or INVALID. It is never automatically “fake” and never automatically “captured”. An asset can lose its manifest for entirely benign reasons, and nothing in this work licenses reading a missing manifest as evidence of intent.

Malicious signer. Cryptographic validity proves that an assertion was signed, not that it is true; a signer willing to assert falsehoods can produce a valid manifest containing them, and no propagation rule can prevent that. The contract constrains conforming operators, not a dishonest issuer. A threat-model matrix pairing attacker capability against contract behaviour is Supplementary Note S3.

4. Formal model

4.1. Evidence records

Let a source asset be partitioned into elements. Each element x carries an evidence record

$$e_x = (P_x, S_x, A_x, L_x), \quad (1)$$

where P_x is provenance, S_x maps each semantically typed support channel to a value in the lifted domain

$$\mathcal{S} = \{\perp_\mu\} \cup [0, 1], \quad \perp_\mu \preceq s \text{ for every } s \in [0, 1], \quad (2)$$

A_x records the scope over which each channel is declared applicable, and L_x is lineage. A channel can be *unavailable*, written $\perp_\mu$, and must then count as weaker than any numeric value it could have carried, otherwise “no emitted channel value is greater” would be silent about a channel that disappears entirely; $\preceq$ on $\mathcal{S}$ is the usual order on $[0, 1]$ extended with $\perp_\mu$ as least element.

Provenance is a set of ancestry atoms drawn from $\{C, G\}$, for captured-derived and generated-derived sources. Mixed ancestry is $\{C, G\}$ and is produced only by union; it is never a primitive. Nothing in the algebra below depends on the size of this set: $\{C, G\}$ is the v1 instantiation, and a deployment carrying finer atoms, per capture device or per generator, inherits the propositions unchanged. The absence of verified evidence is a distinct state $\perp$. It is not an atom, and it is never silently mapped onto C or G . The labels CAPTURED, GENERATED, MIXED and UNVERIFIED are renderings; the normative fields are P , S , A and L , and REAL is not a state anywhere in the model, the implementation or the serialisation.

4.2. The claim order and its meet

Write $\hat{\mathcal{C}} = \{\perp\} \cup (2^{\{C, G\}} \setminus \emptyset)$ for the set of provenance claims, ordered by

$$q_1 \sqsubseteq q_2 \iff q_1 = \perp \text{ or } (q_2 \neq \perp \text{ and } q_1 \supseteq q_2), \quad (3)$$

read “ q_1 is no stronger than q_2 ”: a larger atom set is a weaker claim and $\perp$, which asserts nothing, is the bottom element. It is a policy order: it governs permission to make an exclusive-origin claim, and is not an ordering of truth, credibility or forensic reliability; a known mixture is not the same as uncertainty about origin, and neither captured nor generated means true or false speech. This is a partial order under which two claims always have a greatest lower bound but need not have an

upper bound ($\{C\}$ and $\{G\}$ have none), so the structure is a meet-semilattice rather than a lattice, with meet

$$q_1 \sqcap q_2 = \begin{cases} \perp & \text{if } q_1 = \perp \text{ or } q_2 = \perp, \\ q_1 \cup q_2 & \text{otherwise.} \end{cases} \quad (4)$$

This is a standard construction, an idempotent, commutative, associative combination with an absorbing element, as in provenance semirings (Green et al., 2007) and information-flow lattices (Denning, 1976); we record it rather than derive it.

4.3. The complete-source operator

Let a representation operator T produce output element y from the *complete required source set* $D_y = \{x_1, \dots, x_k\}$. Two notions of dependency diverge for some operators. The *representational* dependency of y is the source content y stands for; the *computational* dependency is the set of source samples that numerically influence its value. The contract governs the representational reading, and the kernel footprint reconciles the two where signal processing makes them bleed together: a resampling filter makes neighbouring content influence y , and since that influence cannot be distinguished from representation at the contract's level of abstraction, the footprint conservatively enlarges D_y to cover it. The one v1 operator for which the two readings diverge completely is amplitude normalisation (Section 9). Write $D_y^\mu = \{x \in D_y : \mu \in A_x\}$ for the sources that declare channel μ applicable. The contract requires:

- (i) **Provenance** $P_y = \bigcap_{x \in D_y} P_x$, that is, the exact union of the required source provenance sets, with $\perp$ absorbing and with no invented ancestry.
- (ii) **Support** if $P_y \neq \perp$, $D_y^\mu \neq \emptyset$, $A_y(\mu) \neq \emptyset$ and every member of D_y^μ supplies a value, then $S_y(\mu) = \min_{x \in D_y^\mu} S_x(\mu)$; otherwise the channel is *unavailable*; without the applicability condition a channel narrowed to an empty scope under (iii) would carry a number applicable nowhere. Requirement (iv) strengthens the condition $D_y^\mu \neq \emptyset$ stated here to $D_y^\mu = D_y$.
- (iii) **Applicability** the *canonical* scope is $A_y(\mu) = \bigcap_{x \in D_y^\mu} A_x(\mu)$, and an empty intersection makes the channel unavailable. A *conforming* operator may emit any $A'_y(\mu) \subseteq A_y(\mu)$, never a broader one; conformance is the containment, the equality names the canonical reduction the reference implementation emits, and statements quantifying over conforming operators, Theorem 1 in particular, are read in that sense.
- (iv) **No partial subsets** a channel is available on the output only if every required source declares it applicable *and* supplies a value: $D_y^\mu = D_y$ is required, not merely $D_y^\mu \neq \emptyset$. A channel unreported by some required source, or not applicable to it at all, is unavailable and is not computed from the remaining sources. The second half is what makes Proposition 2 hold for channels: without it, enlarging D_y by a source that declares an otherwise inapplicable channel would raise that channel from unavailable to a value.
- (iv') **Unverified absorption for channels** if any required source is in state $\perp$, every numeric channel on the output is unavailable.
- (v) **Lineage** $L_y = \bigcup_{x \in D_y} L_x$, identifying the complete required source set. The record is carried per evidence-homogeneous interval, so L_y names the assets required by *that* interval and the interval's own bounds carry the placement.

Requirements (iv) and (iv') are where this model is strictly stronger than its spatial predecessor. (iv) costs expressiveness: sources with different channel schemas yield no channel on a shared interval. (iv') is its companion for $\perp$, since computing a channel from the remaining verified sources would be the partial-subset computation (iv) forbids. Both are tested separately.

4.4. Properties

The guarantee below is that a derived claim can only lose.

Proposition 1 (Non-promotion). *Under (i)–(iv'): $P_y \sqsubseteq P_x$ for every $x \in D_y$, and no atom appears in P_y that appears in no required source; if any required source has $P_x = \perp$ then $P_y = \perp$ and no numeric channel is emitted; and for every channel μ , $S_y(\mu) \preceq S_x(\mu)$ for every $x \in D_y^\mu$ while $A_y(\mu)$ is no broader than any $A_x(\mu)$.*

Proof. By (4), P_y is $\perp$ if any source is, and $\perp \sqsubseteq q$ for every q ; otherwise it is the union of the source sets, which contains each P_x and is therefore $\sqsubseteq P_x$ by (3), and membership of an atom in P_y requires membership in some P_x . Requirement (iv') withholds every channel when $P_y = \perp$. A minimum over a set is bounded above by each member, (ii) refuses to emit unless every applicable source supplies a value, and repeated intersection in (iii) cannot enlarge a set. $\square$

Theorem 1 (Closure under composition). *A finite composition of operators each satisfying (i)–(v), including (iv'), satisfies the same requirements, and the composed claim is no stronger than the meet over the union of all ancestral required source sets.*

Proof. Associativity and commutativity of $\sqcap$ make the composed provenance the meet over the union of the ancestral sets, so no ancestor is dropped or invented; absorption of $\perp$ is preserved inductively because no later operator receives a verified element in place of an unverified ancestor. For a channel μ , each operator either refuses the channel or emits a value bounded by its applicable inputs, and transitivity of $\preceq$ bounds the final value by every applicable ancestor. Repeated intersection cannot broaden a scope, and lineage composes by union of the per-operator source maps. $\square$

Theorem 1 is the standard consequence of combining in a meet-semilattice. The next proposition is not specific to time, since its proof uses only inclusion, union, minimum and intersection, but it is what makes the temporal instantiation implementable against real processing software: more possible source dependencies may weaken an output claim and can never strengthen it, so erring towards a larger footprint errs safely.

Proposition 2 (Footprint monotonicity). *For finite source sets $\emptyset \neq D_y \subseteq D'_y$, the claim the canonical reduction computes from D'_y is no stronger than the claim it computes from D_y , and for every channel μ the value computed from D'_y is $\preceq$ the value computed from D_y in $\mathcal{S}$.*

Proof. Enlarging the index set of a union can only add atoms, and by (3) a superset is a weaker claim; adding a source in state $\perp$ sends the claim to $\perp$, which is weaker still.

For channels the argument runs through (iv), which requires $D_y^\mu = D'_y$ for availability. Suppose μ is available on D'_y . Then every member of D'_y declares μ applicable and supplies a value, so every member of $D_y \subseteq D'_y$ does too, and μ is available on D_y as well; the emitted value is a minimum over the larger set D'_y , which is no larger than the minimum over D_y . Suppose instead μ is unavailable on D_y . Some member of D_y then fails to declare μ applicable, fails to supply a value, or is in state $\perp$, or the scopes over D_y already have an empty intersection; that member lies in D'_y , and an

intersection over more scopes is no larger, so μ is unavailable on D'_y as well, and $\perp_\mu \preceq \perp_\mu$. The remaining case is an added source whose scope empties the intersection in (iii): requirements (ii) and (iii) then withhold the channel, giving $\perp_\mu$, which by (2) is $\preceq$ every element of $[0, 1]$. Every case therefore moves down $\mathcal{S}$.

The condition $D_y^\mu = D_y$ carries the weight here. Under the weaker $D_y^\mu \neq \emptyset$ the statement is false: with $D_y = \{x_1\}$ and μ not applicable to x_1 , enlarging to $D'_y = \{x_1, x_2\}$ where x_2 declares μ with value v would raise the channel from $\perp_\mu$ to v . The hypothesis $D_y \neq \emptyset$ is needed as well: an empty set reduces to $\perp$, and enlarging it to one captured source would strengthen the claim. An output position with no required source is therefore not covered by the proposition, and the implementation marks it unverified. $\square$

Proposition 2 is the practical result of this section: an implementation does not need the exact dependency set of a filtered operator, because a *conservative over-approximation* is always safe and only under-approximation can promote. That converts knowing which input samples every output sample depends on inside a third-party codec into a tractable requirement: declare a set that contains the true one. An analytical radius can establish that; a probe can only test it, and Section 5 says which of the two each operator's declaration rests on.

Signal transparency (P8). The contract must not change the audio. The canonical essence of an asset is the SHA-256 of its decoded stream rendered to signed 16-bit little-endian PCM at the asset's own rate and channel count, and it must be identical whether the EM assertion or the baseline assertion is attached. Whole-file hashes are unusable here because embedding a manifest changes the file by construction (Section 7.5). Equality of this canonical 16-bit rendering is what is established, not equality of every higher-precision decoded representation.

5. Audio instantiation

5.1. Temporal intervals

All positions are sample indices on a named asset and all intervals are half-open $[a, b)$, matching the C2PA convention that “All start times are inclusive of that moment in time, and all end times are, by default, exclusive of it” (Coalition for Content Provenance and Authenticity, 2026a, §18.2.2.3). A source timeline is a gapless ordered partition of an asset into evidence-homogeneous intervals.

The v1 support channel. Version 1 carries one typed channel, **capture-support**, applicable over the single scope **digital-asset**: the degree to which an interval's own record supports a capture claim, on $\mathcal{S}$. It is not a detector output and is not calibrated against any acoustic measurement: what this instantiation demonstrates is that a typed channel aggregates by minimum, absorbs, intersects its scope and survives the transport, not that its numbers carry forensic meaning. The corpus and transport experiments assign 0.90 to captured-derived and 0.10 to generated-derived intervals, so a mixed interval aggregates to 0.10; the enumeration sweeps a grid of values, and the universal statement over $[0, 1]$ comes from the proof. One scope cannot exhibit disjoint or overlapping scopes, nor a channel applicable to one source and not another, the case Proposition 2 turns on; a synthetic battery over 5 scopes covers those (Section 7.1).

An operator emits a set of *map pieces* (Figure 2): output range $[o_a, o_b)$ represents source range $[s_a, s_b)$ on a named asset, with a footprint radius in source samples. Where pieces overlap, as in a mix, the required source set of an output interval is the union over every covering piece. The emitted

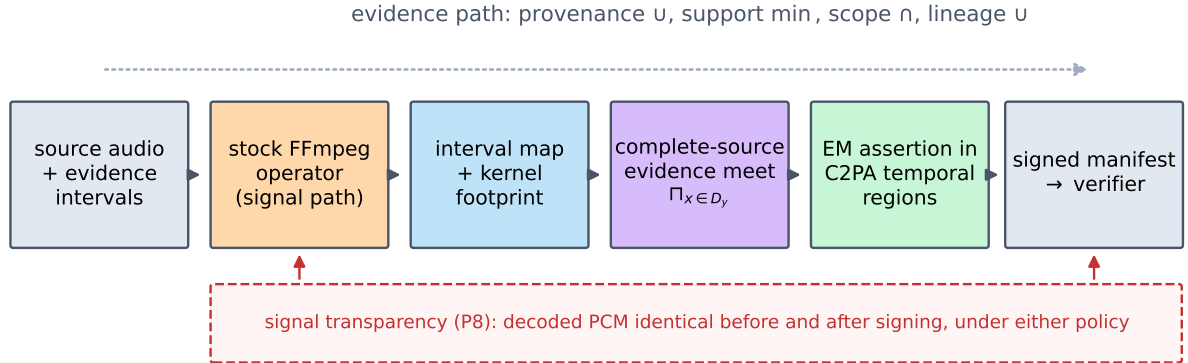


Figure 2: Where the contract sits. Stock FFmpeg performs the transformation; the interval map with its declared kernel footprint determines the complete required source set; the complete-source meet produces the emitted evidence; the result is serialised as C2PA temporal regions of interest and signed. The evidence path never touches the signal path, which is what signal transparency (P8) measures.

output partition is the pull-back of every source evidence boundary across every piece, which gives each emitted interval a stable complete required source set.

5.2. The $v1$ operator set and its kernel footprints

Table 2 gives the eight operators in scope with their mapping and declared footprint. Pitch shifting is not a further primitive: it composes a rate change with a compensating stretch, so Theorem 1 covers it and a named regression test exercises the composite. Denoising, separation and enhancement are out of scope for $v1$ because their dependency structure needs modelling of its own; generative editing differs more basically, since it may introduce new ancestry.

For a polyphase resampler the radius follows from the pinned filter length and rate ratio. At 16 000 Hz `libmp3lame` emits MPEG-2 LSF frames of 576 samples; the radius takes the larger MPEG-1 long window, 1 152, plus the encoder delay, giving 1 728, and one frame of alignment guard. That derivation is necessary but not sufficient: the pinned `libmp3lame` runs with its bit reservoir enabled, which carries bits across frames, so the encoder is stateful in a way no window-and-delay argument captures. The declared 2 304 is therefore a *measured* declaration for this frozen configuration, containing a reach of 1 555 at the operating points probed in Section 7.6; it is not a bound, and the holdout challenge of Section 7.7 finds inputs on which this encoder exceeds it. For overlap-add time stretching the footprint covers the analysis window scaled by the tempo factor; for a frame-granular selector it covers the difference between the exact retained-sample model and the implementation’s frame-aligned cuts. Two experiments check these numbers against software we do not control, and they bound different quantities: Section 7.4 measures temporal mapping deviation, which informs the margin, and Section 7.6 tests whether the *declared* footprint contains the measured source influence. For two operators the measured reach exceeds the analytical support and is contained only by the declaration as a whole, which is why the declaration, not either component, is what an implementation uses. An implementation on a build for which the two quantities have not been validated has no declaration to use, and the contract leaves three courses: refuse the operation, mark the interval unverified, which Section 4 makes absorbing, or widen to whole-asset dependency and pay the dilution of Section 7.9. Emitting the claim an unvalidated declaration would license is the one course forbidden.

Table 2: The v1 operator set. The analytical column is a support radius derived from the pinned algorithm configuration; the margin covers implementation behaviour that the analysis does not capture, such as frame-granular cut points and a correlation search range; the declared footprint is their sum and is the quantity Section 7.6 probes. All values are in source samples for the configurations used in the experiments, so transcoding takes a row per codec and the eight operators occupy nine rows; concatenation assembles the corpus clips rather than transforming them (Section 6.3), leaving the eight transformations Section 7 counts and, among those, the 7 single-source configurations Section 7.6 probes. By Proposition 2 erring large only weakens the emitted claim; erring small is a conformance failure. These values hold for one FFmpeg build and are not portable: on the build used in the independent reproduction of Section 7.12, the MP3 row is under-declared. Section 9 states the normalisation row’s representational reading and its tested alternative.

Operator	Source mapping	Analytical	Margin	Declared	Basis
trim / crop	1:1 on the retained range	0	0	0	exact
concatenate	1:1 per part	0	0	0	exact
resample 16 to 8 kHz	$n_{\text{out}} \cdot f_{\text{in}} / f_{\text{out}}$	33	64	97	polyphase FIR
transcode (MP3)	1:1	1 728	576	2 304	MDCT window + delay; see note
transcode (FLAC)	1:1	0	0	0	lossless
amplitude normalisation	1:1	0	0	0	scalar gain; see note
time stretch 1.10	$t_{\text{src}} = \tau t_{\text{out}}$	529	2 048	2 577	overlap-add window
retained-run selection	explicit retained runs	0	4 096	4 096	frame-granular selector
mix / overlay	1:1 from every covering source	0	0	0	sample-wise sum

Three kinds of footprint evidence. A declared footprint rests on one of three grounds, and the guarantee is as strong as its ground. An *analytical* footprint follows from the algorithm and its pinned configuration: trimming, concatenation, lossless transcoding, fixed-gain scaling, and the polyphase resampler, whose radius is the filter length. A *measured* footprint contains the dependency observed on a stated set of probes; probes can refute it and cannot establish it, because an adaptive operator re-decides when its input changes. The *fallback* is whole-asset dependency, which needs no measurement and gives up local specificity. The lossy encoder and the overlap-add stretcher are adaptive, and a prespecified holdout challenge finds inputs on which both exceed their measured declarations on the reference build itself (Section 7.7); a strict deployment treats them with whole-asset dependency, at the cost Section 7.9 measures. The contract prevents promotion when the declared map contains the relevant source dependencies; the probes measure that condition at stated operating points and do not certify a universal support radius for an adaptive operator.

5.3. The baseline reference policy

The comparison policy throughout is *boundary-only, primary-parent inheritance*: the derived output inherits from its parent ingredient, evaluated at the first and last source elements of the retained region, ignoring kernel footprints. It is a **constructed reference policy**, not a description of any shipping product, and no claim is made here about what any named tool does. Its motivation is the specification text: a derived asset’s relationship to its input is **parentOf**, “a derived asset or asset rendition of this ingredient” (Coalition for Content Provenance and Authenticity, 2026a, Table 10, §18.16.3), and the minimal way to inherit from a parent over a retained region is to read its endpoints. Since the specification does not define the reduction, every implementation

that derives an asset supplies one, deliberately or by default, and endpoint inheritance is one such rule. The baseline isolates one consequence of that underspecification without attributing it to any implementation; how deployed software behaves would need a survey this paper does not contain.

The baseline combines two independent shortcuts, boundary-only inheritance and blindness to the kernel footprint, so the experiments also evaluate the two intermediate policies that remove one shortcut each (Section 7.3). That ablation keeps the comparison from resting on the choice of baseline: a reader who disputes the composite policy can read the result off the intermediate arm.

5.4. Signed transport

Evidence is carried as a namespaced C2PA assertion, `io.github.1d-coda.emaudio.evidence`, with one entry per output evidence interval. Each entry gives a temporal region of interest as an `npt` range (Coalition for Content Provenance and Authenticity, 2026a, §18.2.2.3), the sample-exact bounds alongside it because `npt` strings are lossy at sample resolution, the provenance atom list (`null` for unverified, never an atom), the available support channels, their applicability scopes, and the lineage set. Derivation records the source as an ingredient with relationship `parentOf`; the specification requires a corresponding `c2pa.opened` action carrying a hashed-URI reference to that ingredient assertion (Coalition for Content Provenance and Authenticity, 2026a, §15.11.3.2), which the claim generator inserts. Composition records each source as a `componentOf` ingredient carrying its full source manifest, with a `c2pa.placed` action referencing each component’s ingredient assertion (Coalition for Content Provenance and Authenticity, 2026a, §18.16.3), so the transport exercises exactly the composition vocabulary whose missing aggregation rule motivates the paper (Section 7.8). The full schema, with a worked manifest, is Supplementary Note S4.

The transport is a bridge, not a re-implementation: signing and validation are performed by the official `c2patool`, so the transport guarantees are the tool’s. The choice of C2PA is not load-bearing; the same assertion can be carried in a detached signed sidecar, and C2PA is used because it is the deployed standard and tests the mapping against a real validator.

6. Methods

6.1. Study design

The experiments separate three questions that are easy to conflate: whether the implementation realises the stated guarantees (a mathematical question, answered by exhaustive enumeration and by differential testing); whether a competing metadata rule produces counterexamples on adversarial and on realistic input (a mechanism question, answered by frozen fixtures and a labelled corpus); and what the contract costs (an engineering question, answered by timed repetitions). Exact algebraic properties and finite-state classifications are reported as deterministic counts and are not assigned p -values, and Table 8 states, for each class of claim, which evidence is used for it and what that evidence does not establish. Fixture failure frequencies characterise the frozen fixture family and are a **mechanism stress rate**: they are not, and must not be read as, an estimate of how often any deployed system promotes provenance.

6.2. Deterministic fixtures

Experiment A enumerates every source word of length 1 to 8 over $\{C, G, \perp\}$, giving 9 840 words, and pushes each through up to ten operator configurations. Two configurations require a minimum word length: the inner trim needs at least three elements and retained-run selection at least two.

So the 3 single-element words receive eight configurations, the 9 two-element words nine, and the remaining 9 828 words the full ten, giving 98 385 cases. Each case is checked against the property suite and against a closed-form oracle that recomputes the expected state from the operator’s nominal source ranges without the evidence algebra; zero-footprint operators must match it exactly, and kernel operators may only be weaker.

Experiment B freezes 10 000 timelines of 64 source intervals each, generated from a fixed seed, in which the majority of intervals are captured and one to four *non-boundary* intervals are replaced by generated or unverified evidence. Each is pushed through composition chains of depth 1 to 5 and, separately, through each operator alone. Because the anomalies are deliberately placed away from the boundaries, the baseline’s failure rate on this family is high by construction. To make that explicit, the experiment includes a control arm in which anomaly positions are drawn uniformly over all positions, for which the baseline’s promotion probability has the closed form $\binom{n-2}{k} / \binom{n}{k}$ for k anomalies of a single type over n positions.

6.3. Mixed-origin audio corpus

600 clips are built deterministically, each of the form captured | generated | captured with sample-exact ground truth. Captured segments are crops of LibriSpeech `dev-clean` (Panayotov et al., 2015), CC BY 4.0; generated segments are produced locally by eSpeak NG formant synthesis from phrases written for this study. No speaker is identified and no voice is cloned. Segment durations are matched so that the generated region is a modest interior interval.

Because the evidence algebra reads no acoustic feature, the result cannot in principle depend on signal character. A robustness arm demonstrates that rather than leaving it to argument, rebuilding 50 clips with a neural TTS (Piper, a VITS-architecture engine with a voice trained on the public-domain LJSpeech corpus) and overlaying 50 clips with locally generated pink noise mixed at a volume setting of -18 dB over every ancestry boundary, a volume setting rather than a measured signal-to-noise ratio.

Concatenation and every subsequent transformation are performed by **stock FFmpeg through its command line, with no project code in the signal path**: the artefact that does or does not exhibit promotion is produced by software the author does not control, and only the evidence policies are under study.

6.4. Metrics

We report promotion counts and rates, lineage omissions, unverified-to-verified promotions, support promotions, conformance failures, decoded-essence mismatches, manifest validation outcomes, model-versus-implementation sample deviations, runtime and metadata size. We deliberately do not report accuracy, precision, recall, F_1 or AUC: no classifier is evaluated. Timing is reported as medians with interquartile ranges over 30 repetitions on a single machine, an Apple M3 Max with 14 logical cores and 36 GiB, running macOS 26.5.2, Python 3.11.15, FFmpeg 9.0.1 and `c2patool` 0.27.2; every version is recorded in `results/PREFLIGHT.txt`, regenerated on each run. Each repetition processes 30 clips in one process with no affinity or power control and no warm-up, which moves the median by 1.22%.

6.5. Reproducibility and AI use

Every number in this manuscript is emitted by code into `results/machine_readable/`, converted into a LaTeX macro by a generator script, and used in the text through that macro; no result is

typed by hand. Every figure is rendered from the same files. `run_all.sh` exits non-zero if any conformance check fails. Generative-AI tools were used as programming and review assistants, under the declaration at the end of this paper.

7. Results

7.1. Exhaustive conformance

Over 9 840 source words and 98 385 operator cases, the property suite and the closed-form oracle together evaluated 984 213 checks with 0 failures, including 363 explicit composition cases. Every named regression test passes, across the contract and review-regression suites together: 49 of 49, with 0 failures (Table 6).

That enumeration carries one channel over one scope, which is what the audio instantiation uses, so it cannot reach the applicability cases of (iii). A separate synthetic battery covers those: over 5 scopes, chosen so that pairs are disjoint, nested and partially overlapping, it enumerates 441 source-set enlargements and checks each against Proposition 2, with 0 violations.

An earlier version of requirement (iv) withheld a channel only when a required source declared it applicable and reported no value, leaving a source to which the channel was not applicable at all to be silently skipped. Under that rule the enumeration still passed all 984 213 checks, because in v1 every non- $\perp$ source declares the one channel; the battery fails it in 80 of 441 cases, each an enlargement that raised a channel from unavailable to a value. The monotonicity check had the same blind spot, comparing a channel only when it was present on both sides. Both are fixed and the battery runs in the pipeline.

During development the suite failed on the overlay operator, where the implementation evaluated overlapping map pieces independently and emitted a captured claim and a generated claim over the same output interval instead of one mixed claim; that was a defect in our own code, fixed by making the output partition global across pieces. Passing checks are limited evidence; these regressions, and the mutation tests that reject an inexact reduction, show that the suite is sensitive to these particular defects.

7.2. Adversarial timelines

On 10 000 frozen timelines, boundary-only inheritance promoted the provenance claim on 9 455 timelines (94.5%) at composition depth 1 and on 9 455 at depth 5; complete-source propagation produced 0 promotions summed over every depth, 0 unverified-to-verified promotions, 0 lineage omissions and no support promotion. Boundary-only inheritance omitted required lineage on 10 000 timelines at depth 1 and reported 7 222 unverified spans as verified.

The rate does not rise with depth, and that is the finding: the loss is established at the first derivation and is irreversible, because once the interior evidence is discarded no downstream operator can recover it (Figure 3A). Across the 9 operator configurations applied singly, baseline promotion ranged from 94.3% to 100.0% and complete-source propagation produced 0 promotions (Figure 3C). In the control arm, with anomaly positions uniform over all positions, measured baseline rates agreed with the closed-form prediction to within 0.0070 in absolute rate (0.70 percentage points) in every cell (Figure 3B). The adversarial arm’s near-unity rate is thus a property of the construction, arithmetically predictable, and carries no information about deployed software; the informative quantity is the exact zero on the other side.

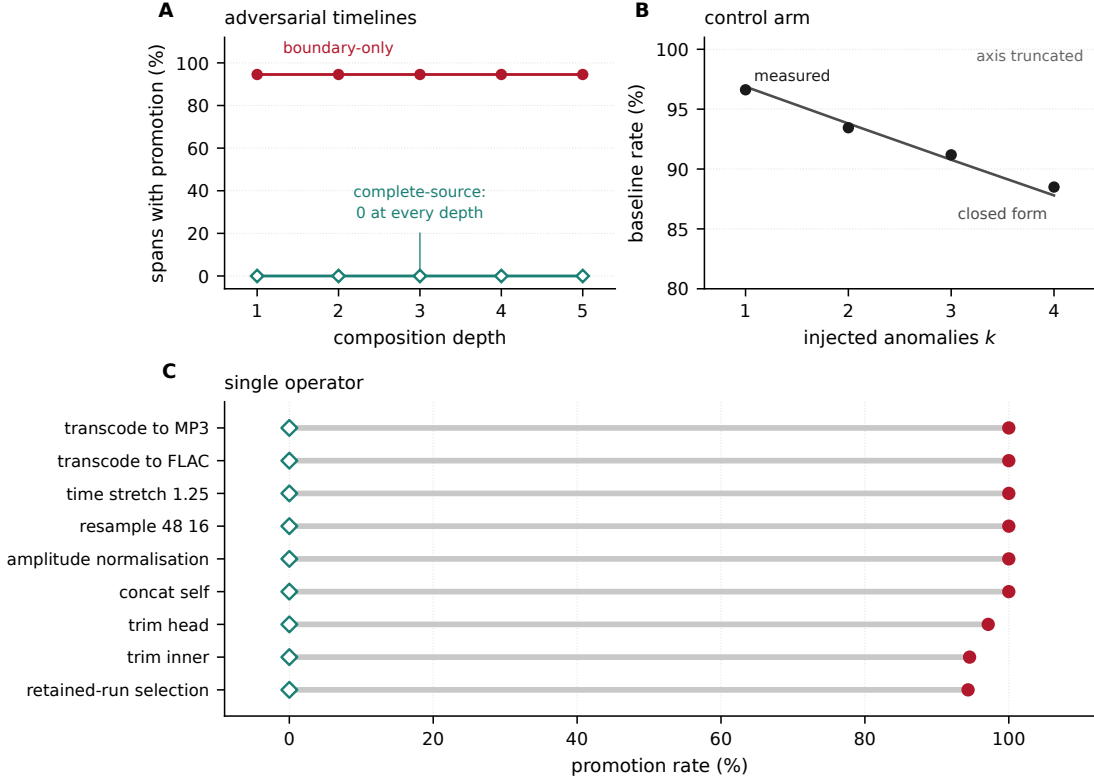


Figure 3: Boundary-only versus complete-source inheritance on 10 000 deterministic timelines. (A) Promotion does not grow with composition depth, because the loss is established at the first derivation and no downstream operator can recover evidence already discarded. (B) Control arm with anomaly positions uniform over all positions, measured against the closed form $\binom{n-2}{k}/\binom{n}{k}$, on an axis truncated to 80–100% to expose prediction error. (C) Each operator applied alone: filled circles are boundary-only, open diamonds complete-source, which is zero for every operator. Percentages throughout quantify mechanism stress on a frozen fixture family and are not real-world prevalence.

7.3. Which shortcut matters: a policy ablation

The baseline combines two shortcuts, and removing them one at a time over 10 000 timelines per arm gives an unambiguous answer (Table 3). In the *interior* arm the anomaly lies inside the nominal represented range, so only the inheritance axis can see it: both boundary-only policies promoted on every timeline (10 000 and 10 000) and both complete-source policies on none. In the *footprint* arm the anomaly lies just outside the nominal range but inside the declared kernel footprint: both footprint-blind policies promoted on every timeline (10 000 and 10 000) and both footprint-aware policies on none. Neither axis is redundant and neither is sufficient, which is the argument for specifying them together.

7.4. Mixed-origin audio through stock FFmpeg

On 600 clips, complete-source propagation recovered the evidence interval structure exactly, every boundary at sample resolution, for 600 clips, and propagated the supplied generated-source annotation with its exact bounds for 600, with no detector, no training and no acoustic feature. The ground truth is constructed rather than discovered: the question is whether evidence originating at the capture and generation steps *survives* a chain of representation-only transformations, not whether it could be inferred from the signal, and nothing here supports a claim that generated audio

Table 3: Policy ablation over 10 000 timelines per arm. The interior arm places the anomaly inside the nominal represented range; the footprint arm places it outside that range but inside the declared kernel footprint. Each shortcut fails exactly one arm, and only the full contract clears both.

Policy	Inheritance	Kernel footprint	Interior anomaly	Footprint anomaly
B0	boundary-only	no	10 000 (100%)	10 000 (100%)
B1	boundary-only	yes	10 000 (100%)	0 (0%)
B2	complete-source	no	0 (0%)	10 000 (100%)
B3	complete-source	yes	0 (0%)	0 (0%)

can be identified without evidence. The robustness arm behaves identically: 50 of 50 neural-TTS clips and 50 of 50 noise-overlaid clips recovered their full interval structure, with 0 and 0 promotions; under the noise overlay the captured segments correctly become MIXED while the generated segment stays GENERATED, since union adds no atom. Boundary-only inheritance promoted the whole-asset claim to captured-only on 600 clips and omitted required lineage on 600; the complete-source policy produced 0 promotions and 0 omissions.

Across 8 transformations and 4 800 runs (Figure 4; Table 6), boundary-only inheritance promoted on 3 600 runs and complete-source propagation on 0, with 0 lineage omissions. Two transformations produced *no* baseline promotion, and we report that as readily as the rest: overlay declares the generated source as a separate top-level ingredient, so a boundary-only reading still sees both, and retained-run selection happened to retain runs whose boundaries fell inside the generated region.

The interval model was validated against FFmpeg’s actual output on every PCM run and on a 100-clip subset for the coded containers. The largest deviation between predicted and actual output sample count was 681 samples for retained-run selection, 471 for time stretching, 31 for MP3 and 0 for resampling; for all transformations the declared mapping margin covered it. That measurement compares output *lengths* only: it does not establish interior alignment, which the probes of Section 7.6 measure, nor the *support containment* Proposition 2 requires, because an operator can emit exactly the predicted number of samples while depending on sources outside the declared footprint; containment is tested directly in Section 7.6. Re-running each transformation and comparing decoded essence gave 0 mismatches, so the processing path is deterministic on this machine. The zero promotion count follows from Proposition 1 once the required source set is correct, and on overlay, during development, it did not; the falsifiable measurements are the mapping deviation and margin coverage here, and support containment in Section 7.6.

7.5. Signed transport, derivation and provenance loss

Across 120 signed round-trips over `wav`, `flac` and `mp3`, 40 clips per container, signing succeeded and validation reported the trusted state under the declared local anchor for 120 assets, with 0 validation failures. The EM assertion returned byte-equivalent from the validator for 120 assets. After derivation with stock FFmpeg, re-signing with the parent as an ingredient, and re-validation, 120 derived assets validated and 120 recorded the `parentOf` relationship.

Signal transparency holds exactly. Decoded essence was identical before signing, after signing with the complete-source assertion and after signing with the baseline assertion, with 0 mismatches against the pre-signing essence and 0 between policies. The whole-file hash changed in 120 of 120 cases, which is why the comparison is defined on decoded PCM; a clean-clone rerun makes the same point, every essence hash agreeing and no signed-file hash, each signature carrying its own timestamp.

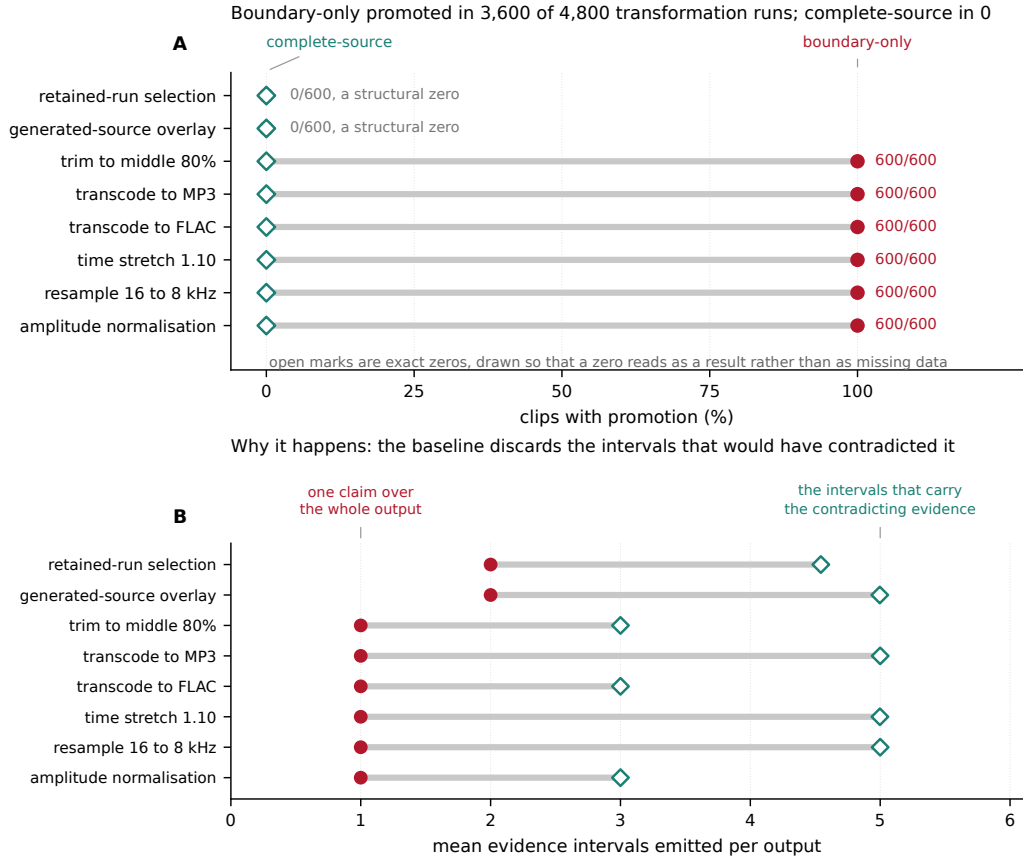
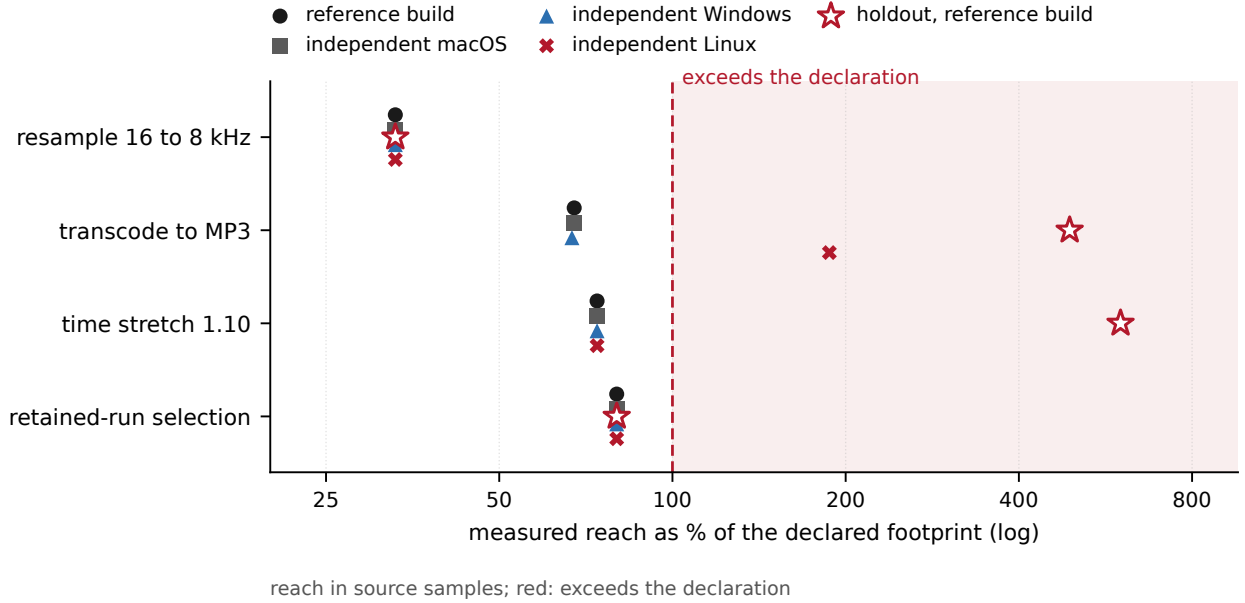


Figure 4: Promotion by transformation over 600 mixed-origin clips processed entirely by stock FFmpeg, 4800 transformation runs in all. Each row is a pair of marks: a filled circle for the boundary-only reference policy and an open diamond for complete-source propagation, with counts written directly rather than read off the axis. Boundary-only promoted in 3600 of 4800 runs and complete-source in 0. Generated-source overlay and retained-run selection produce no baseline promotion either, for the structural reasons given in Section 7.4, and are marked as such rather than left to look like missing rows. (B) The mechanism behind panel A, measured in the same runs: the mean number of evidence intervals each policy emits per output. Boundary-only inheritance emits one interval for most transformations, a single claim spanning the whole output, while complete-source propagation emits three to five. The promotion in panel A is a consequence of that collapse, since an interval that no longer exists cannot carry the evidence that would have contradicted the endpoints. The baseline is a constructed reference policy, not the behaviour of any named deployed product, and these are counts over a frozen corpus rather than a prevalence estimate.



	declared	reference	ind. macOS	ind. Windows	ind. Linux	holdout
resample 16 to 8 kHz	97	32	32	32	32	32
transcode to MP3	2,304	1,555	1,555	1,541	4,317	11,292
time stretch 1.10	2,577	1,906	1,906	1,906	1,906	15,482
retained-run selection	4,096	3,277	3,277	3,277	3,277	3,277

Figure 5: Kernel-support reach against the declared footprint. Each mark is the furthest any one source sample influenced the output beyond its nominal source range, as a percentage of the declaration on a logarithmic axis; the dashed line is the declaration, and marks to its right exceed it. Circles, squares, triangles and crosses are the containment probes of Section 7.6 on the reference build and the three independent runs (Section 7.12); stars are the maximum over the holdout challenge on the reference build (Section 7.7). The table gives each reach in source samples. Zero-footprint operators are omitted. Probes can refute a declaration and cannot establish one.

Under the five provenance-loss conditions over 60 clips, a valid manifest yielded MIXED in 60 cases, a removed manifest yielded UNVERIFIED in 60 cases, an asset modified after signing yielded INVALID in 60, and re-encoding without manifest preservation yielded UNVERIFIED in 60. No condition other than a valid manifest ever yielded CAPTURED, and there were 0 violations of the classification requirement. We do not claim that stripping is detected as malicious intent; the result is that the classification degrades in the conservative direction.

7.6. Kernel-support containment

Proposition 2 needs the declared required-source set to contain the actual dependency set, a statement about which source samples influence which output samples, not about how many output samples there are. We test it directly on the 7 single-source configurations; concatenation and overlay take several sources and have no temporal kernel to under-declare. For each configuration, each of 4 signal contexts and each of ten probe positions, two 16 384-sample sources are built that are identical except at one sample, where one carries an added impulse; both go through the same stock FFmpeg command and both outputs are decoded, so the entire difference between the paired

outputs is attributable to that one sample. Containment holds when the probe position lies inside the declared required-source range of every output sample whose value it changed, on 16-bit PCM at a threshold of one least significant bit.

Two properties of the probe had to be got right, and getting them wrong first showed why: against silence every operator appears to map one sample to one sample, and an earlier fixed-amplitude impulse, roughly two hundred times the carrier in the low-level context, replaced an adaptive operator’s input rather than perturbing it, so both the lossy encoder and the stretcher re-decided globally. The impulse is now a fixed multiple of each context’s peak. The 4 contexts, all generated arithmetically (`experiments/support_containment.py`), are a steady low-frequency tone; the same tone at one part in a hundred, where a lossy encoder allocates fewest bits; sharp onsets with exponential decay, which drive an overlap-add stretcher hardest; and six inharmonic partials, where masking is most active. Lossy transcoding spreads a single source sample across up to 1 823 output samples at every position, resampling across up to 33, and overlap-add stretching is strongly position-dependent, reaching up to 2 023 at the 7 probes inside an overlap region and one sample elsewhere, which is why the test is run against the worst case rather than the median.

Across 280 probes, 52 005 output samples were influenced and 0 fell outside the declared support (Table 4; Figure 5); every context produced influence for every operator, so no row passes vacuously. That zero was not free. A shorter probe passes trivially for the large-footprint operators because a declared footprint of 2 577 samples spans a whole short source; lengthening it made the test able to fail, and it did. Retained-run selection failed: a frame-granular selector emitted a frame beginning 3 277 source samples before a retained run’s start, well beyond the margin declared at the time. That is an under-declaration, the one direction Proposition 2 does not forgive; the band was raised to 4 096, above the measured reach, and the operator now contains with a margin of 819. We report this rather than quietly shipping the corrected number because it shows the declarations are measured rather than asserted. The bands are also reasonably tight, which matters because a larger one would be equally safe and more dilutive: resampling reached 32 against 97, MP3 1 555 against 2 304, and time stretching 1 906 against 2 577.

That outcome accepts an influenced output sample beyond the model’s extent by checking it against the last piece’s whole declared range. Counted strictly, as a sample the emitted map does not cover, 6 such samples appear, in retained-run selection and time stretching, so the strict outcome is 6 rather than 0: an extent shortfall of the mapping, not a support exceedance. Interior alignment, the offset between where the model places a probed source sample and where its influence peaks, reached 205 samples for retained-run selection and 401 for time stretching, inside their declared margins, and 1 for resampling; a correct output length had not shown either.

7.7. Holdout challenge to the declarations

Passing those probes shows that each declaration contains what was measured at the operating points used while it was being set. A finite probe family can refute a declaration and cannot establish one for an adaptive operator, so we challenged the declarations with inputs that played no part in setting them, fixed before the run (`experiments/footprint_holdout.py`): 6 signal contexts, 2 of them new (white noise and a speech-like voice); 4 absolute impulse amplitudes from 256 to 12 000, all within 16-bit range; and 8 positions disjoint from those above, 192 probes per operator and 1 152 in total, on the reference build.

Resampling, retained-run selection and fixed-gain normalisation showed no exceedance, and resampling’s reach stayed at 32 against its analytical 97. The two adaptive operators did not hold. MP3 exceeded its declaration in 4 of 192 probes, all on the low-level carrier at the largest amplitude,

Table 4: Kernel-support containment. Probed as Figure 5 describes. The measured reach is the furthest any influence extended beyond the nominal, footprint-free source range, so containment compares two concrete quantities rather than resting on an assertion: it holds exactly when the reach does not exceed the declaration. Headroom is the difference, so a small headroom means a tight declaration and a large one a conservative one. [†] Probed with a fixed gain: the content footprint of a sample-wise scaling. Estimating the gain from the input adds a whole-asset control dependency, measured in Section 7.7.

Operator	Declared	Measured reach	Influenced samples	Headroom	Outside
amplitude normalisation [†]	0	0	40	0	0
resample 16 to 8 kHz	97	32	1 131	65	0
retained-run selection	4 096	3 277	28	819	0
time stretch 1.10	2 577	1 906	10 971	671	0
transcode to FLAC	0	0	40	0	0
transcode to MP3	2 304	1 555	39 763	749	0
trim to the middle 80%	0	0	32	0	0

reaching 11 292 source samples; time stretching exceeded in 14, on the low-level carrier and on the held-out speech-like voice, at every amplitude from 256 upward, reaching 15 482. A large impulse on a quiet carrier makes the encoder re-allocate bits and the stretcher’s correlation search re-align globally. That is a dependency of the output on the source sample, and a change of operating mode is no ground for excluding it. These two declarations therefore characterise the tested operating points on the build that produced them, and a strict deployment uses whole-asset dependency for these operators (Section 5), whose cost Section 7.9 reports.

Normalisation was run in two modes. With a fixed gain no exceedance occurred. With the gain estimated from each input, as peak normalisation does, 84 of 192 probes influenced output samples across the asset, up to 15 600 source samples away: a control dependency through the gain, not a content one. The contract labels origin by the content an output sample represents and so excludes control dependencies by definition; the measurement shows that this is a semantic choice, not an empirical property of the operator.

7.8. C2PA-native heterogeneous composition

The strongest form of the standards argument is executable. For 30 fixtures, a captured source and a generated source are each signed with their own `digitalSourceType`; stock FFmpeg concatenates them; the composition is signed with both sources as `componentOf` ingredients carrying their full manifests, a `c2pa.placed` action referencing each, and the EM assertion computed over the two component timelines; it is then derived across the ancestry boundary and re-signed with `parentOf`.

`c2pa.mixed` is defined for “multiple, previously placed (via `c2pa.placed` actions) audio ingredients (e.g., stems, vocals, drums, bass, etc) [that] are combined and optionally transformed” (Coalition for Content Provenance and Authenticity, 2026a, Table 8). Our fixture concatenates in sequence rather than summing, and we read that as placement rather than mixing, a *documented profile decision* rather than a normative one, since the specification settles neither that nor `c2pa.remixed` (Coalition for Content Provenance and Authenticity, 2026a, Table 8). `c2pa.mixed` presupposes `c2pa.placed`, so the placed actions are required under any reading, and the choice does not affect the contract, which reads the ingredient graph and the interval map, not the action label.

All 30 compositions validated as trusted under the declared anchor, with both `componentOf` ingredients recorded in 30 cases and both `c2pa.placed` references resolved in 30; the EM assertion returned byte-equivalent in 30; the whole-asset aggregate was MIXED with both intervals individually

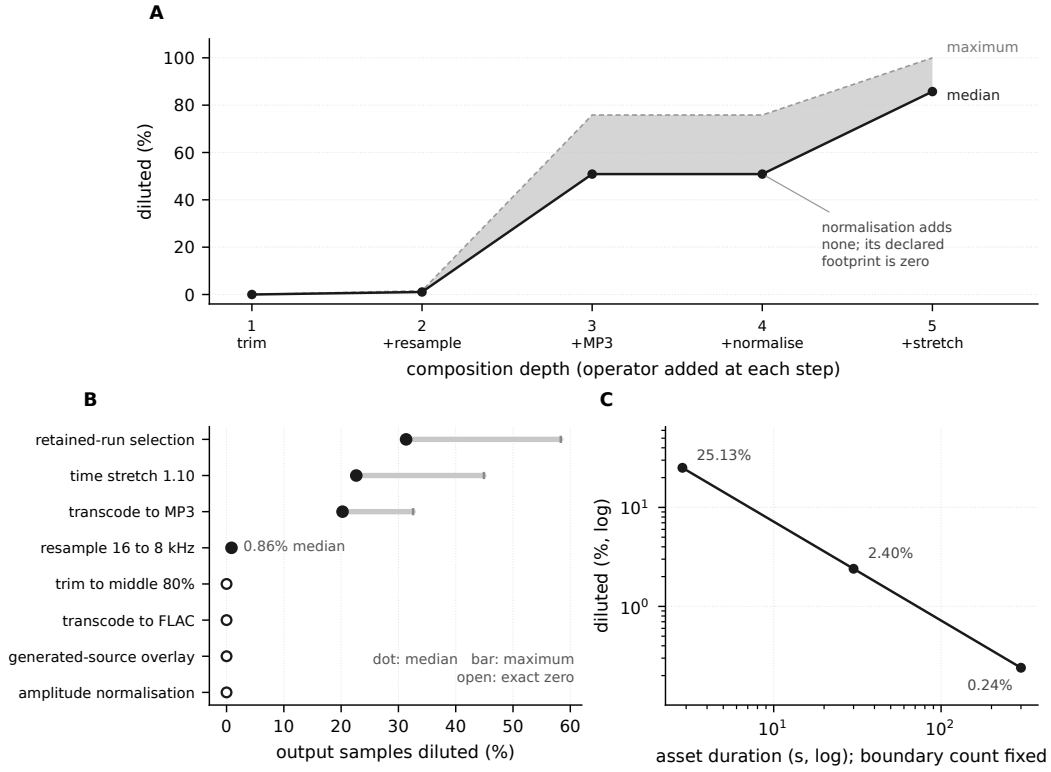


Figure 6: What the declared footprints cost, as the fraction of output samples whose claim is weaker than over the nominal map. (A) Through a composition chain, with the operator each step adds; the shaded band runs from median to maximum over clips and is not an uncertainty interval. Normalisation adds nothing: its content footprint is zero. (B) Each transformation alone over 600 clips: circles are medians, bars reach the maximum, open circles are exact zeros. (C) One depth-three chain at three durations with the boundary count fixed, on logarithmic axes. Dilution weakens a claim and never strengthens one.

recoverable in 30; and the derived asset stayed MIXED and trusted in 30 of 30 derivations, with 0 essence mismatches. This is the object the vocabulary-versus-reduction distinction describes: the component histories are all present in the signed graph, and the aggregate interval claim exists only because the EM assertion computes it. Nothing in the validation pipeline required it to be conservative. A representative manifest is frozen as a fixture in the repository.

7.9. The cost of conservatism: claim dilution

Zero promotion is only half of a useful result, because safety alone is cheap: declaring the whole asset as every sample’s dependency is safe under Proposition 2 and gives up all local specificity. We therefore measure what the declared footprints cost. The *dilution fraction* of a derived output is the fraction of its samples whose footprint-aware evidence record is strictly weaker, in claim, channel or support value, than the record computed over the nominal represented content. It measures conservatism added relative to the nominal map, not a comparison with a known true codec dependency.

Measuring this caught a second defect in our own implementation, reported for the same reason as the first. Footprint widening was originally applied at the granularity of the emitted interval partition, so an interval whose widened range touched a neighbouring ancestry took the weakened claim over its *entire* extent, and a single resampling reported a dilution fraction of one: safe, and

uselessly conservative. The fix refines the output partition at the pull-backs of each source boundary shifted by $\pm$ the footprint, confining dilution to footprint-wide bands at ancestry boundaries. The two-language differential had not caught this because its fixtures used source intervals narrower than every footprint, a geometry in which the coarse and exact treatments coincide; the oracle corpus now includes a wide-interval battery, and the fixed implementation agrees with the per-sample oracle on all 10 860 cases.

After the fix (Table 5; Figure 6), 4 of 8 transformations dilute nothing, lossless transcoding among them, and overlay because its nominal required-source set already contains every covering source. The remainder dilute the footprint-wide neighbourhood of each ancestry boundary: median dilution is 0.86% for resampling, 20.24% for MP3, 31.34% for retained-run selection and 22.64% for time stretching, with a single-transformation worst case of 58.33%. Each band has a *fixed radius in samples* and the clips average 2.85 s, so the same bands on a one-hour asset would occupy a fraction three orders of magnitude smaller: the fraction shrinks with duration, the absolute band does not.

The whole-asset comparator, one claim over the output computed from every source interval of the asset, promotes nowhere (0 promoted samples) and dilutes every sample of every corpus transformation (median 100.00%; 96.99% for overlay, whose nominal map is already partly mixed), against a median of at most 31.34% under the declared footprints. That difference is what the local footprints buy: the same non-promotion with the interval structure kept. It is also the cost of the strict profile for the two adaptive operators of Section 7.7.

Composition is where conservatism bites. Along the chain trim, resample 16 to 8 kHz, transcode to MP3, normalise, time stretch by 1.10, deliberately front-loading a rate change so that the large bands are scaled by it, median dilution reaches 85.74% by depth 5 with a worst case of 99.99%; depth 4 equals depth 3 to the last digit because the operator added there, normalisation, has a zero footprint. Whether a rate change rescales upstream bands is not isolated by this design, whose only rate change occurs before any upstream band exists. The chain is a model of how declared footprints compound: its MP3 and stretch stages run at 8 kHz, configurations Section 7.6 did not probe, so it does not show that those declarations contain the chained audio’s dependency. Holding a routine three-operator chain fixed (trim, normalise, MP3) and varying only duration, dilution is 25.13% at the corpus median length, 2.40% at 30 s and 0.240% at five minutes: each tenfold increase in duration divides the fraction by ten, as fixed-width bands predict when the number of boundaries is held fixed, which it is here and would not be for an arbitrary chain. Dilution therefore dominates only where a deep chain of large-footprint operators meets a short asset with many ancestry boundaries, which is what this measurement deliberately constructs; the quantity a deployment should carry is the band radius of Table 2, not a percentage measured on 2.85-second clips. This is the price of declaring conservative bounds instead of knowing exact codec dependencies, and it can be bought back without losing safety: any tighter map that remains an over-approximation reduces dilution and, by Proposition 2, cannot introduce promotion.

7.10. Two-language differential

The Python reference implementation works with interval algebra over pulled-back source boundaries; the Node.js oracle evaluates evidence per output sample by brute force and run-length-encodes the result, with no interval algebra at all. The frozen corpus spans footprints wider than every source interval and footprints far narrower, the regime added after the dilution experiment (Section 7.9). It is stored as `fixtures/oracle_cases.jsonl`, one case per line carrying the interval model and the source evidence in a form that names neither implementation, and the oracle reads it as a separate process. On 10 860 cases the two agreed on interval structure, provenance state,

Table 5: Claim dilution: the fraction of output samples whose footprint-aware evidence is strictly weaker than the evidence over the nominal represented content, per transformation over 600 clips and along a composition chain over 200 clips. This is the price of the declared footprints, which combine analytical support with implementation margin, when the nominal map is exact.

Transformation	Clips	Median dilution	Max dilution	Clips affected
amplitude normalisation	600	0.00%	0.00%	0
overlay a generated source	600	0.00%	0.00%	0
resample 16 to 8 kHz	600	0.86%	1.39%	600
retained-run selection	600	31.34%	58.33%	600
time stretch 1.10	600	22.64%	44.92%	600
transcode to FLAC	600	0.00%	0.00%	0
transcode to MP3	600	20.24%	32.56%	600
trim to the middle 80%	600	0.00%	0.00%	0
composition depth 1	200	0.00%	0.00%	—
composition depth 2	200	1.06%	1.59%	—
composition depth 3	200	50.87%	75.84%	—
composition depth 4	200	50.87%	75.84%	—
composition depth 5	200	85.74%	99.99%	—
same depth-3 chain, 2.85 s asset	1	25.13%	—	—
same depth-3 chain, 30 s asset	1	2.40%	—	—
same depth-3 chain, 300 s asset	1	0.24%	—	—

lineage set and channel availability with 0 disagreements, and the maximum absolute difference in any support value was 0. Both implementations are author-written, so this is a differential test against transcription and interval-arithmetic error, not evidence of independence from the author (Section 9).

7.11. Cost

Complete-source bookkeeping cost a median 1.42 ms per audio-minute against 0.42 ms for boundary-only, a ratio of $3.37\times$ computed from the unrounded medians, and 0.114% of the FFmpeg time it accompanies (Table 7). Signing took a median 32.7 ms per asset (IQR 24.6–53.7) and validation 29.1 ms (IQR 22.5–40.0). Manifest overhead was 14 932 B per asset, of which the EM assertion was 1 333 B; the fixed per-asset component (certificate chain, COSE structure, hard binding) dominates, and the variable part grows linearly in emitted evidence intervals at 333 B and $35.2\ \mu\text{s}$ per interval, measured up to 256 with no departure from linearity. Cost therefore scales with how often ancestry changes along the timeline, not with duration. It also scales with how far the codec smears that ancestry: the assertion is 1 333 B in `wav` and 1 334 in `flac` against 2 114 in `mp3`, whose footprint splits it into more intervals. An implementation that wants a ceiling on the count may coarsen the partition, merging adjacent intervals and emitting the meet over the merged range: each sample then carries the union of the merged required source sets, which by Proposition 2 can only weaken its claim, at the dilution cost of Section 7.9. The byte figure is deterministic and reproduces exactly; the microsecond figure is wall clock on one interpreter and one machine.

Of these figures the ratio travels and the milliseconds do not, and the mechanism is structural: both arms are timed in the same process on the same machine, so thermal state, clock frequency or competing load moves them together and cancels in the quotient. Re-running the benchmark as 9 independent processes gives the absolute cost a coefficient of variation of 2.79% against 2.21% for the ratio, with the per-process rows and their load averages in Supplementary Note S10; across sittings

Table 6: Main results. Deterministic fixture rates characterise the frozen fixture family and are a mechanism stress rate, not a population prevalence estimate.

Check	Cases	Boundary-only	Complete-source
<i>A Exhaustive finite-state conformance</i>			
Source words over $\{C, G, \perp\}$, length ≤ 8	9,840	—	—
Operator cases	98,385	—	—
Contract checks	984,213 checks, 0 failed		
Named regression tests	49 tests, 49 passed, 0 failed		
<i>B Deterministic adversarial timelines (10 000 frozen fixtures, 64 intervals)</i>			
Chain depth 1	10,000	9,455 (94.5%)	0
Chain depth 3	10,000	9,455 (94.5%)	0
Chain depth 5	10,000	9,455 (94.5%)	0
Lineage omissions (depth 1)	10,000	10,000	0
Unverified $\rightarrow$ verified (depth 1)	—	7,222	0
<i>C–D Mixed-origin audio corpus, stock-FFmpeg processing</i>			
Exact propagation of constructed interval evidence	600	—	600
amplitude normalisation	600	600 (100.0%)	0
overlay a generated source	600	0 (0.0%)	0
resample 16 to 8 kHz	600	600 (100.0%)	0
retained-run selection	600	0 (0.0%)	0
time stretch 1.10	600	600 (100.0%)	0
transcode to FLAC	600	600 (100.0%)	0
transcode to MP3	600	600 (100.0%)	0
trim to the middle 80%	600	600 (100.0%)	0
<i>H Two-language differential oracle</i>			
Frozen cases	10,860	—	0 disagreements

Table 7: Cost, as medians over 30 repetitions on one machine. These are measurements of a Python reference implementation, not language-independent constants.

Quantity	Median	IQR
EM bookkeeping	1.416 ms / audio-minute	1.369–1.980 ms / audio-minute
Boundary-only bookkeeping	0.420 ms / audio-minute	0.411–0.429 ms / audio-minute
EM / boundary-only ratio	3.37×	—
EM as a fraction of FFmpeg time	0.114%	—
C2PA signing	32.7 ms / asset	24.6–53.7
C2PA validation	29.1 ms / asset	22.5–40.0
Manifest size overhead	14,932 B / asset	14,932–14,933
EM assertion	1,333 B / asset	1,333–1,334
EM assertion marginal cost	333 B / evidence interval	—

on this machine the absolute figure has moved by about a third while the ratio stayed inside its band. On a quiet machine with little common mode to cancel the quotient can carry slightly more scatter than either arm, and we have observed both cases. Cancelling variation within a machine is not cancelling the difference between two, so the cross-machine question is settled by measurement in Section 7.12: the absolute figure differs by 15% and the ratio by 4.8%. The milliseconds are order-of-magnitude evidence that the bookkeeping is cheap, not reproducible constants.

Table 8: Claim–evidence boundary and what transfers. Each row names a class of claim, the evidence used for it, what that evidence does not establish, and whether the independent reproduction of Section 7.12 reproduced it. Evidence from one row is never used to authorise a claim in another.

Claim class	Evidence used here	Not established by it	Transfers?
Operator conformance	Propositions 1 and 2 and Theorem 1; exhaustive enumeration; 441-case scope battery; 10 860-case two-language differential	That any deployed system conforms; the prevalence of non-conformance	Yes: proved from the requirements; 885 828 checks of the suite he was given, 0 failures; the current 984 213 reproduced exactly by the Windows run of tag v1.0.7; scope and oracle cases reproduced exactly
Mechanism on constructed fixtures	10 000 frozen adversarial timelines with a closed-form control arm	Any population rate; the behaviour of any named product	Yes: 9 455 baseline promotions against 0, reproduced exactly
Mechanism on realistic audio	600 labelled mixed-origin clips processed by stock FFmpeg	Balanced coverage of speakers, languages, codecs or editing styles	Yes: 0 promotions against 3 600, reproduced exactly
Interval-model fidelity	Predicted versus actual output sample counts on every PCM run and a 100-clip coded subset, against declared margins	Kernel-support containment, a different property; the margins on other codecs, builds or configurations	No : retained-run selection deviation rises from 681 to 12 505 samples against a 4 096-sample margin
Kernel-support containment	280 probes: ten positions in each of 4 contexts across 7 operator configurations, on decoded PCM at one least significant bit	Containment at untested positions, signals, codecs or configurations; the operators as configured here, not the operator class; which configuration produced a given asset, which the assertion does not record	No : the MP3 declaration is under-sized on the other build, reach 4 317 against 2 304
Signed transport and signal transparency	120 round-trips through the official tool over wav, flac and mp3; decoded-PCM SHA-256 identity under both policies	Conformance-program trust; security of the C2PA cryptographic layer; perceptual equivalence; bit-exactness across decoders; any container the signing tool accepts but this experiment did not carry	Yes: identical before and after signing on both machines
Consumer-side verification	Per-interval lineage and the ingredient graph, both returning byte-equivalent through the official tool (Sections 7.5, 7.8)	Detection of promotion: the assertion records an operator, not the chain, so a consumer cannot recover the required source set	No for detection; the schema omits it, not the build
Runtime cost	30 timed repetitions on one declared machine	Cost in a compiled implementation or at broadcast scale	No for milliseconds, 15% apart and 13.2% dispersed within one sitting; approximately for the ratio, 4.8% apart on one pair of machines
Conservatism cost	Per-sample dilution against constructed exact ground truth	Dilution against the true, unknowable dependencies of a third-party codec	Yes: reproduced exactly

Table 9: The reference machine and the three third-party runs. *MP3 reach* is the measured kernel reach in source samples against a declared footprint of 2304. The reference and two of the runs use FFmpeg 9.0.1; those two are different builds of that one version and do not agree, which is why the declaration is stated per build. Full environments and per-operator detail are Supplementary Note S8.

Run	Platform	FFmpeg	Verifier outcome	MP3 reach
Reference (this paper)	macOS 26.5.2	9.0.1	—	1 555
D. A. Balderrama-Alvarez	Linux 6.6.87.2	8.0.1-3ubuntu2	REPRODUCTION INCOMPLETE; 192 deterministic differences	4 317
B. C. Guerra Flores	macOS 26.3.1	9.0.1	Exit zero; 17 differences, all timings	1 555
D. A. Balderrama-Alvarez	Windows 11	9.0.1-full_build- www.gyan.dev	69 deterministic differences	1 541

7.12. Independent reproduction

The classification of outputs was fixed in code and shipped before any third party ran it (Section 9), so what follows is a protocol executed rather than a comparison arranged after the fact.

D. A. Balderrama-Alvarez (Universidad de Sonora) ran the package for release tag v1.0.0 twice on his own machine (Table 9). He is not an author, took no part in designing the contract or the experiments, and received the package without any result files: this is independence of execution, not of implementation (Section 9). The first run was incomplete because two dependencies were absent from the package’s instructions, since fixed; the second, reported here, produced the same values for the two findings below, which rules out run-to-run variation on his hardware. The package he was given, tag v1.0.0, is earlier than the released v1.2.0; neither finding’s value changed between the two, which we checked rather than assumed, and what did change is the size of the conformance suite, as recorded below. His raw outputs are released unmodified in **results/independent/**.

tools/verify_reproduction.py returned **REPRODUCTION INCOMPLETE** with the 192 differences reported below, and **run_all.sh** ended in **RUN FAILED** for two packaging defects that touch no reported number (Supplementary Note S7). Both are printed because Section 9 makes that tool the arbiter precisely so its verdict cannot be adjusted after the fact.

What reproduced. Of the 13 result files carrying a deterministic claim, 10 reproduced with no deterministic difference, across an operating system, processor architecture, Python version and FFmpeg major version that share nothing with the reference machine. Every headline outcome is among them, including the 80 cases the superseded rule failed, and Table 8 reports them row by row. Decoded-essence identity before and after signing held for all 120 round-trips; the digests differ between machines, as a different FFmpeg decodes to different bytes, and the claim is identity within a machine.

What did not, and why it is a result. 192 deterministic differences remain, in 3 files. 3 of them are the size fields of Experiment A and are not a failing check: his package’s suite had 885 828 checks and every one passed, and the released suite has 984 213 because the structural checks were added after his run; the later Windows run of tag v1.0.7 reproduces the 984 213 exactly, and every deterministic output of that tag is identical to that of the evaluated release, v1.2.0. The remaining 189 fall in two files and trace to the same cause, the one Section 9 names in advance: the kernel footprints of Table 2 are declarations calibrated against one FFmpeg build, and his is a different one.

3 of the 4 operators that declare a non-zero footprint reproduced their measured reach exactly across the two builds; the remaining 3 declare no footprint and measure no reach on either machine, so their agreement carries no information. The supplement gives the per-operator comparison in full.

For the MP3 encoder, the declared footprint of 2304 source samples was adequate on the reference build, where the measured reach was 1555. On his build the reach is 4317, exceeding the declaration by 2013, and 110 output samples fall outside the declared support, all in the transient probe context. This is a failure of one declared number, not of Proposition 2, which covers enlargement and not under-sizing, and its consequence is exactly the failure this paper is about: a source’s genuine influence reaches samples the assertion does not attribute to it, which is promotion. An over-declared footprint costs dilution; an under-declared one costs the guarantee, and a reader on an untested FFmpeg build should assume the same until the calibration says otherwise.

For retained-run selection, the deviation between the modelled temporal mapping and the one FFmpeg performs rises from 681 samples on the reference build to 12505 on his, against a declared margin of 4096. The median deviation is 9105.5, a systematic offset rather than an outlier, and `aselect` retains 16384 samples on his build where the reference retains 13312. His containment arm for this operator nonetheless reported no sample outside the declared support: mapping and support are two quantities validated by two experiments, and only the first failed.

Two builds, one version number. Two later runs on the released package separate the version number from the environment around it (Table 9). The Guerra Flores run is the first by a third party to reproduce every deterministic output of the package it ran; the Windows run’s 69 differences all lie in the MP3 kernel measurements and the time-stretch model deviation, on which the two builds also disagree, 471 against 463 samples, both inside the 2048-sample margin.

What that supports is narrower than it may look. The version number is 9.0.1 in three of the four rows and the reach takes two values, so pinning a version is not a substitute for re-calibrating; machine and operator do not move it, since the two runs that agree share neither. Operating system, architecture and build differ together, so we claim only that the number is not fixed by the version, and these runs were made to test the package rather than as an experiment. The declaration of 2304 samples contained every measurement in this group, with 0 and 0 probes outside.

We report the two findings on the Linux build as findings rather than as reproduction failures, and we do not re-declare the numbers to make them pass: widening them until his build conforms would produce a table adequate for two builds and tested against neither. What the reproduction establishes is the scope of Table 2: a calibration for a stated configuration, for which the re-calibration of Section 9 is a prerequisite and the two-pass tool that performs it is the artefact that transfers.

That outcome is consistent with the broader forensic-tool literature without validating anything here: Nordvik and Axelsson (2022) showed that a shared specification does not guarantee consistent interpretation across implementations, while Külper et al. (2026) showed that forensic outputs can drift substantially across software versions. Our own runs add a narrower observation: matching FFmpeg version strings do not guarantee identical calibrated behaviour across builds. The contract is portable because it is a proof; the calibrated constants are not, and are not assumed to be.

Cost, and which figure travels. His bookkeeping cost is 1.62 ms per audio minute against 1.42 on the reference machine, a difference of 15%; the ratio to the baseline is 3.54 against 3.37, a difference of 4.8%. Within his sitting, across 9 processes, the absolute figure has a coefficient of variation of 13.2% against 1.6% for the ratio, the shape Section 7.11 predicts: the quotient helps in proportion

818 to the common-mode variation there is to cancel. This is one pair of machines, so it shows the ratio
819 surviving a change of hardware once rather than bounding how far it can move.

820 8. Discussion

821 *What is established.* That a complete-source rule is implementable over media time against real
822 processing software; that it eliminates promotion on every fixture and corpus clip tested, where a
823 minimal boundary-only policy does not; that it can be carried in an existing signed standard without
824 altering the audio; and that its cost is a fraction of a percent of the processing it accompanies.
825 An independent run on unrelated hardware returned the headline contract and corpus outcomes
826 unchanged; of its 192 deterministic differences, 3 are the size of a suite that grew after the run and
827 189 trace to declarations his FFmpeg build does not satisfy (Section 7.12). Independent execution
828 shows those counts are not artefacts of the reference machine; that they are properties of the contract
829 follows from Propositions 1 and 2 and Theorem 1. Table 8 states in one place what transfers:
830 everything derived from the requirements, and nothing derived from a particular build, a boundary
831 that runs through Table 2.

832 *What is not.* No claim is made that any deployed system promotes provenance; the baseline is a
833 constructed reference policy and the fixture rates are a mechanism stress rate. Nor is the algebra
834 new: the non-amplification property is a meet in a semilattice, familiar from provenance semirings
835 (Green et al., 2007) and information-flow models (Denning, 1976; Goguen and Meseguer, 1982), and
836 can equally be read as a “no write up” discipline for evidential authority.

837 *Relation to the standard and to adjacent work.* The layer addressed here is the reduction over a
838 vocabulary that already preserves what the problem needs, and it requires no specification change
839 (Section 5.4). It has not been put to the C2PA working group and no endorsement is claimed;
840 standardisation of the reduction, rather than of this label, would be the useful outcome. Detectors,
841 watermarks and chunk-localised binding (Ono, 2026) answer different questions and compose with
842 the contract, which is most useful where detectors are weakest, on short interior edits in genuine
843 recordings, and useless on media that arrive with no evidence; on the specification’s security we cite
844 a public analysis (Golaszewski et al., 2026) rather than assert.

845 *Attack surface.* The contract closes one provenance-laundering pathway, a mixed-origin asset cleaned
846 by passing through a benign conforming operator, and does nothing against a malicious signer, a
847 compromised key or an actor who strips provenance and starts a dishonest chain (Section 3).

848 *What a consumer can check.* How much of the rest a consumer can check is a property of the
849 transport, and the schema settles it. Each interval carries its atoms and the lineage naming its
850 required sources (Section 5.4), and the ingredient graph carries each component’s manifest, both
851 returning byte-equivalent through the official tool (Sections 7.5, 7.8); a consumer walking that
852 graph can detect *invented* ancestry, an atom the emitted claim asserts and no named source holds,
853 which is the second half of Proposition 1. Detecting promotion is the harder direction and v1 does
854 not support it: the assertion records the operator that produced the output and its parameters,
855 not the chain, so where a derivation composed several operators the required source set cannot be
856 recovered from it, and a claim stronger than the meet over whole sources is also what a legitimate
857 trim produces (Table 8). Carrying the map, or the operator chain that determines it, would supply
858 it and is the schema change we would make first. The omission surfaced when we tried the check.

No consumer check reaches an operator that misdeclares its own operation or footprint, whose effect is the promotion Section 7.12 measures at 110 output samples and which no consumer can tell from a build mismatch: the dishonest signer of Section 3 rather than the negligent one addressed here.

Deployment, and what a reader may conclude. The contract deploys as a library beside the processing step, whose caller already knows the parameters that fix the interval map and footprint, so nothing touches the signal path; interoperation is demonstrated with the official `c2patool` only. For whoever reads the output, an interval labelled MIXED is a statement about the declared sources and not about the audio, and an asset carrying a valid manifest without an evidence assertion says only that its producer did not implement the contract. Absence is not a finding.

Extension. Nothing in Section 4 refers to audio; another medium costs its interval algebra and footprint table, which we have not built. Spatial extents make the union of Proposition 1 run over regions, and geometric warps are the two-dimensional form of the position-dependent footprint that time stretching already previews, which is an expectation and not a result.

9. Limitations

Independence. The oracle and the evidence policies are author-written; the differential tests transcription and interval-arithmetic error, not author independence. Two components are outside our control by design: all processing is stock FFmpeg and all signing the official `c2patool`. Three runs by two other people are reported in Section 7.12; none is independent of the design of the experiments, only of their execution. One reproduced every deterministic output exactly on a build matching the reference, which shows the pipeline deterministic rather than differing in ways we find convenient to explain; the other two, on differing builds, found the declarations that do not transfer. A fourth person reported two portability defects on Windows without completing a run, which improved the software and is not a reproduction. A third-party reimplementations remains outstanding and is the stronger check. What it would run against is the 10860 frozen cases of `fixtures/oracle_cases.jsonl`, not the enumeration's 98385 operator cases, which are generated against our own objects and cannot leave the language; the second implementation of Section 7.10 already consumes that file across a process boundary. Disagreement would be the more informative outcome, since it would locate an ambiguity in the specification.

What an independent reproduction has to compare. The criteria were fixed in `tools/verify_reproduction.py` and shipped before any third party ran it: deterministic outputs are held to exact equality and any difference exits non-zero, environment-dependent outputs are reported without failing. Fixing the comparison in code stops the criteria being adjusted once a result is in hand. The membership of each class and the protocol a further reproduction should follow are Supplementary Note S7.

One validator, not an interoperability result. Every round-trip is produced and checked by the official `c2patool` under a locally generated credential and trust anchor, so *trusted* here means trusted under that anchor. A second API, `c2pa-python`, also recovered the assertion on every asset, but it binds the same Rust core, which rules out a CLI-specific artefact and nothing further. No independently implemented consumer has read the assertion, so semantic interoperability is not established; the assertion is designed to degrade safely, an unaware consumer being left with the standard ingredient graph, but that is an intention, not a tested outcome.

900 *Corpus.* 600 clips from one read-speech corpus and one formant synthesiser at 16 000 Hz is a
901 mechanism corpus, not a balanced design over speakers, languages, conditions, codecs or editing
902 styles. It supports the claim that the mechanism is real and the rule works on realistic audio, not
903 any rate claim about audio in general. The propagation rule is independent of acoustic content, since
904 the contract never reads the audio; the footprint declarations are not, because adaptive and lossy
905 operators depend on the signal (Section 7.6), which is why they hold only at the tested configurations
906 and operating points. The axes that can change the outcome are therefore operator and codec
907 configuration, signal context for adaptive operators, and ancestry geometry. Production material
908 at other rates, channel counts and codecs would further validate the footprint layer; it is not a
909 requirement of the contract, and more codec configurations are the direction Section 7.12 shows to
910 be load-bearing.

911 *Footprints are declarations.* The kernel footprints are declarations for the exact configurations used
912 here: analytical for the linear operators, measured for the two adaptive ones, and the holdout
913 challenge of Section 7.7 finds inputs on which the measured ones are exceeded on their own build.
914 A different FFmpeg build, resampler setting, codec or tempo factor requires re-declaration and re-
915 testing, and Proposition 2 guarantees only that erring large is safe. Re-declaration needs two passes,
916 one per quantity of Section 5, and neither substitutes for the other: output length can be exact
917 while the dependency is wider than declared. Both are in `tools/calibrate_footprint.py`, which
918 exits non-zero whenever a declaration fails to contain its measured reach. The support pass probes
919 ten prespecified positions in each of 4 contexts, not an exhaustive sweep, and adaptive operators
920 re-decide when the input changes enough, so the declared footprints characterise dependency around
921 the tested operating points, not a universal support radius; where no local radius can be supplied,
922 whole-asset dependency remains safe at the dilution cost of Section 7.9. Two of the declarations
923 shipped here sit below the tool’s recommended margin while containing their reach, and are marked
924 as such rather than quietly raised.

925 *Content and control dependency.* Peak normalisation is where the two diverge completely: each
926 output sample represents one source sample, and the gain that scales it depends on the whole
927 signal, which the holdout challenge measures directly. The contract labels origin by represented
928 content, so the footprint is zero and the gain is a control dependency outside the label. That is
929 a semantic choice the measurement cannot justify, and the same choice governs the encoder’s bit
930 allocation and the stretcher’s alignment search. A deployment that counts control dependencies
931 gives such operators whole-asset dependency; the maximally conservative reading of normalisation
932 is implemented and tested, and pays dilution without gaining promotion safety.

933 *Replay, identity and truth.* A microphone recording a loudspeaker playing synthetic speech produces
934 audio that is genuinely captured at the sensor layer; every capture-provenance system has this
935 limitation, and the contract would propagate a captured claim, correct at the layer it models. The
936 honest response is layering, with capture mechanism, generation ancestry and speaker identity
937 as separate channels; version 1 stays at the digital-asset and source-lineage level and detects no
938 re-recording. Identity is not provenance, and neither is truth: a captured recording can contain a
939 lie, and the contract says nothing about semantic content.

940 *From a machine-readable claim to an examiner’s reading.* The motivation drawn from [Andersen](#)
941 [et al. \(2025\)](#) is a human-factors result: examiners erred on how a tool presented data. What this
942 work supplies is a machine-readable claim, and whether an examiner reads MIXED correctly and acts

943 differently for it is untested here. Closing that distance needs a study with examiners, not another
944 conformance suite.

945 10. Conclusion

946 A representation-only transformation may change, compress, aggregate or reorganise content, but
947 it must not silently increase the evidential authority of the output relative to the complete source
948 material it represents. In audio that principle has an exact form: provenance is the union over the
949 complete required source set, support the minimum, applicability the intersection, lineage the union,
950 and an unverified source interval is absorbing. The algebra is not new. This work provides the
951 executable temporal contract, the kernel-footprint treatment that makes it implementable against
952 real codecs, a signed transport inside an existing standard, and measurements showing that the
953 rule eliminates promotion on every fixture and clip tested, leaves the decoded audio untouched
954 and costs a fraction of a percent of the processing it accompanies. An independent reproduction
955 returned those results unchanged and found two build-specific declarations that do not transfer,
956 which locates the part of this work that has to be re-measured before it is reused. Non-promotion is
957 proved from the requirements; the audio instantiation delivers it where the declared map contains
958 the real dependencies, which holds analytically for the linear operators and, for the encoder and the
959 stretcher, only at the operating points tested: a holdout challenge on the reference build exceeds
960 both, and a strict deployment gives them whole-asset dependency. It does not detect deepfakes, does
961 not establish that speech is truthful, and a valid signature proves only that an assertion was signed.

962 Data availability

963 The mixed-origin corpus is generated deterministically by committed code from LibriSpeech
964 `dev-clean` (Panayotov et al., 2015), licensed CC BY 4.0, fetched by a checksum-pinned script, and
965 from locally generated eSpeak NG output. The robustness arm also uses a Piper neural text-to-
966 speech voice trained on the public-domain LJ Speech corpus, fetched and checksum-verified rather
967 than committed, and pink noise generated locally, seeded per clip. No new human participants
968 were recruited; existing public speech data were used, and no proprietary or login-walled data. No
969 voice of any identified living person is cloned. Licences, versions, URLs, access dates, digests and
970 redistribution clauses are recorded in `DATA_LICENSES.md`.

971 Code availability

972 The reference implementation, the second-language oracle, every experiment, every frozen
973 fixture and every machine-readable result are released under the MIT licence with pinned tool
974 versions and one-command reproduction at <https://github.com/1D-Coda/em-audio>, public at
975 submission rather than on acceptance. The unmodified outputs of the independent reproductions of
976 Section 7.12, with their preflight reports, pipeline logs and classified comparisons, are released in
977 `results/independent/`, `results/independent_mac/` and `results/independent_windows/`. The
978 repository ships a frozen copy of the released results in `results/reference/`, the comparison target
979 when `tools/verify_reproduction.py` runs from a downloaded archive, and a container image and
980 continuous verification that re-measure the declarations on further builds, author-run evidence not
981 reported here. The evaluated state is release tag `v1.2.0`, the identifier to resolve; the commit hash
982 in `results/PREFLIGHT.txt` is necessarily the parent of the commit that stores the report, and the

983 report says so. The audio artefact is deposited in Zenodo with version and concept DOIs before
984 submission, cited in the Data availability statement, and the repository tag and the Zenodo version
985 resolve to the same evaluated state.

986 **CRediT authorship contribution statement**

987 **Alejandro Urias:** Conceptualization, Methodology, Software, Formal analysis, Validation,
988 Investigation, Visualization, Writing – original draft, Writing – review and editing.

989 **Funding**

990 This research did not receive any specific grant from funding agencies in the public, commercial,
991 or not-for-profit sectors.

992 **Declaration of competing interest**

993 The author declares no competing financial or personal interests. No commercial product
994 is experimentally evaluated or compared in this work; named forensic products appear only in
995 describing findings reported in the cited peer-reviewed literature, and no claim is made that any
996 named product implements the constructed reference policy or exhibits provenance promotion.
997 The comparison policy is a constructed reference policy derived from the C2PA specification’s
998 own `parentOf` definition; all processing is stock FFmpeg and all signing the official `c2patool`,
999 neither authored or controlled by the author; and a non-conformance found in the author’s own
1000 implementation is reported in Section 7.1 rather than quietly fixed.

1001 **Declaration of generative AI and AI-assisted technologies in the manuscript preparation** 1002 **process**

1003 During the preparation of this work, the author used Claude Opus 5 and Opus 5.5 (Anthropic)
1004 and ChatGPT (OpenAI) to improve language clarity and readability, and as programming assistants
1005 on the verification and reproducibility tooling. Every reported value was obtained by executing
1006 the committed experiments, and the author verified the code that produced it. After using these
1007 services the author reviewed and edited the content as needed and takes full responsibility for the
1008 content of the publication.

1009 **Acknowledgements**

1010 We thank Daniel A. Balderrama-Alvarez (Universidad de Sonora, ORCID 0009-0002-5180-
1011 0406) for the reproductions reported in Section 7.12, for reporting their failures without adjusting
1012 them, and for the portability defects his Windows 11 runs found in code paths hosted continuous
1013 integration had not exercised; Miguel Arroyo (Nagoya Industries Promotion Corporation, ORCID
1014 0009-0008-8423-8345) for testing the reproduction software on Windows 11 and reporting two further
1015 defects invisible to hosted continuous integration, in tool location and in directory removal under
1016 Windows file locking, both fixed in the archived release; and Brenda Cecilia Guerra Flores (ORCID
1017 0009-0000-8932-683X) for the reproduction reported in Section 7.12, the first by a third party to
1018 return every deterministic output unchanged, on hardware and tool versions she chose. None of
1019 them had any part in the design of the contract, the experiments or the classification of outputs,
1020 and none is an author.

References

- Andersen, D.B., Sunde, N., Porter, K., 2025. Tool induced biases? Misleading data presentation as a biasing source in digital forensic analysis. *Forensic Science International: Digital Investigation* 52, 301881. doi:[10.1016/j.fsidi.2025.301881](https://doi.org/10.1016/j.fsidi.2025.301881).
- Burri, X., Casey, E., Bollé, T., Jaquet-Chiffelle, D.O., 2020. Chronological independently verifiable electronic chain of custody ledger using blockchain technology. *Forensic Science International: Digital Investigation* 33, 300976. doi:[10.1016/j.fsidi.2020.300976](https://doi.org/10.1016/j.fsidi.2020.300976).
- Coalition for Content Provenance and Authenticity, 2026a. C2PA technical specification, version 2.4. https://spec.c2pa.org/specifications/specifications/2.4/specs/C2PA_Specification.html, accessed 19 August 2026.
- Coalition for Content Provenance and Authenticity, 2026b. Use of Content Credentials to identify synthetic and non-synthetic content. C2PA implementation guidance. Published July 2026; <https://c2pa.org/resources/>, accessed 20 August 2026.
- Denning, D.E., 1976. A lattice model of secure information flow. *Communications of the ACM* 19, 236–243. doi:[10.1145/360051.360056](https://doi.org/10.1145/360051.360056).
- European Parliament and Council of the European Union, 2024. Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence (artificial intelligence act). Official Journal of the European Union, L series. URL: <http://data.europa.eu/eli/reg/2024/1689/oj>. eLI identifier; Article 50 transparency obligations applicable from 2 August 2026.
- Fu, X., Wang, Z., Zhang, Y., Chen, Y., 2026. Trust the voice, hide the source: Anonymous provenance for verifiably edited audio. *Cryptology ePrint Archive*, Paper 2026/1308. <https://eprint.iacr.org/2026/1308>; preprint, not peer reviewed at the time of writing.
- Gerhardt, M., Cuccovillo, L., Aichroth, P., 2024. Audio provenance analysis in heterogeneous media sets, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, Workshop on Media Forensics*, pp. 4387–4396. https://openaccess.thecvf.com/content/CVPR2024W/WMF/html/Gerhardt_Audio_Provenance_Analysis_in_Heterogeneous_Media_Sets_CVPRW_2024_paper.html, accessed 20 August 2026.
- Goguen, J.A., Meseguer, J., 1982. Security policies and security models, in: *1982 IEEE Symposium on Security and Privacy*, pp. 11–20. doi:[10.1109/SP.1982.10014](https://doi.org/10.1109/SP.1982.10014).
- Golaszewski, E., Krawetz, N., Sherman, A.T., Ziegler, E., Matukumalli, S.K., Yus, R., Kegley, C.L., Barthel, M., Bowman, W., Barot, B., Kullman, K., 2026. Verifying provenance of digital media: Security analysis of C2PA and its implementation. *Cryptology ePrint Archive*, Paper 2026/804. <https://eprint.iacr.org/2026/804>; preprint, not peer reviewed at the time of writing.
- Green, T.J., Karvounarakis, G., Tannen, V., 2007. Provenance semirings, in: *Proceedings of the Twenty-Sixth ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems (PODS '07)*, pp. 31–40. doi:[10.1145/1265530.1265535](https://doi.org/10.1145/1265530.1265535).
- Hargreaves, C., Nelson, A., Casey, E., 2024. An abstract model for digital forensic analysis tools – a foundation for systematic error mitigation analysis. *Forensic Science International: Digital Investigation* 48, 301679. doi:[10.1016/j.fsidi.2023.301679](https://doi.org/10.1016/j.fsidi.2023.301679).
- Külper, M., Lambertz, M., Rybalka, M., 2026. Do you dare to try test-driven forensics? Increasing trust in desktop forensics with ADARE. *Forensic Science International: Digital Investigation* 57, 302115. doi:[10.1016/j.fsidi.2026.302115](https://doi.org/10.1016/j.fsidi.2026.302115).
- Nemecek, A., He, H., Cheng, G., Ayday, E., 2026. Authenticated contradictions from desynchronized

- provenance and watermarking. arXiv:2603.02378. <https://arxiv.org/abs/2603.02378>; preprint, not peer reviewed at the time of writing.
- Nordvik, R., Axelsson, S., 2022. It is about time—do exFAT implementations handle timestamps correctly? *Forensic Science International: Digital Investigation* 42–43, 301476. doi:[10.1016/j.fsidi.2022.301476](https://doi.org/10.1016/j.fsidi.2022.301476).
- Ono, T., 2026. MerkleSpeech: Public-key verifiable, chunk-localised speech provenance via perceptual fingerprints and Merkle commitments. arXiv:2602.10166. <https://arxiv.org/abs/2602.10166>, submitted 10 February 2026; preprint, not peer reviewed at the time of writing.
- Panayotov, V., Chen, G., Povey, D., Khudanpur, S., 2015. LibriSpeech: An ASR corpus based on public domain audio books, in: 2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 5206–5210. doi:[10.1109/ICASSP.2015.7178964](https://doi.org/10.1109/ICASSP.2015.7178964).
- Pham, L., Lam, P., Tran, D., Tang, H., Nguyen, T., Schindler, A., Skopik, F., Polonsky, A., Vu, C., 2025. A comprehensive survey with critical analysis for deepfake speech detection. *Computer Science Review* 57, 100757. doi:[10.1016/j.cosrev.2025.100757](https://doi.org/10.1016/j.cosrev.2025.100757).
- Qamhan, M., Alotaibi, Y.A., Selouani, S.A., 2023. Transformer for authenticating the source microphone in digital audio forensics. *Forensic Science International: Digital Investigation* 45, 301539. doi:[10.1016/j.fsidi.2023.301539](https://doi.org/10.1016/j.fsidi.2023.301539).
- San Roman, R., Fernandez, P., Défossez, A., Furon, T., Tran, T., Elshahar, H., 2024. Proactive detection of voice cloning with localized watermarking, in: Proceedings of the 41st International Conference on Machine Learning (ICML), pp. 43180–43196. URL: <https://proceedings.mlr.press/v235/san-roman24a.html>. peer-reviewed conference version; preprint arXiv:2401.17264.
- Urias, A., 2026. Evidence-monotone terrain representation framework (EMTRF): A provenance-conserving operator contract for derived terrain geometry. Companion work on a different medium, under review elsewhere; its own reproducibility archive, not this paper’s, is concept DOI <https://doi.org/10.5281/zenodo.21992294>.
- Wang, X., Delgado, H., Tak, H., Jung, J.w., Shim, H.j., Todisco, M., Kukanov, I., Liu, X., Sahidullah, M., Kinnunen, T., Evans, N., Lee, K.A., Yamagishi, J., et al., 2026. ASVspooF 5: Design, collection and validation of resources for spoofing, deepfake, and adversarial attack detection using crowdsourced speech. *Computer Speech and Language* 95, 101825. doi:[10.1016/j.csl.2025.101825](https://doi.org/10.1016/j.csl.2025.101825).
- World Wide Web Consortium, 2013. PROV-DM: The PROV data model. W3C Recommendation. <https://www.w3.org/TR/prov-dm/>, accessed 19 August 2026.